Incorporating Administrative Data in Survey Weights for the Basic Monthly Current Population Survey

by

Jonathan Eggleston
U.S. Census Bureau

Yarissa Gonzalez
U.S. Census Bureau

Carl Lieberman
U.S. Census Bureau

Tim Trudell
U.S. Census Bureau

John Voorheis
U.S. Census Bureau

CES 24-02 **January 2024**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact Sean Wang, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov. To subscribe to the series, please click here.

Abstract

Response rates to the Current Population Survey (CPS) have declined over time, raising the potential for nonresponse bias in key population statistics. A potential solution is to leverage administrative data from government agencies and third-party data providers when constructing survey weights. In this paper, we take two approaches. First, we use administrative data to build a non-parametric nonresponse adjustment step while leaving the calibration to population estimates unchanged. Second, we use administratively linked data in the calibration process, matching income data from the Internal Return Service and state agencies, demographic data from the Social Security Administration and the decennial census, and industry data from the Census Bureau's Business Register to both responding and nonresponding households. We use the matched data in the household nonresponse adjustment of the CPS weighting algorithm, which changes the weights of respondents to account for differential nonresponse rates among subpopulations.

After running the experimental weighting algorithm, we compare estimates of the unemployment rate and labor force participation rate between the experimental weights and the production weights. Before March 2020, estimates of the labor force participation rates using the experimental weights are 0.2 percentage points higher than the original estimates, with minimal effect on unemployment rate. After March 2020, the new labor force participation rates are similar, but the unemployment rate is about 0.2 percentage points higher in some months during the height of COVID-related interviewing restrictions. These results are suggestive that if there is any nonresponse bias present in the CPS, the magnitude is comparable to the typical margin of error of the unemployment rate estimate. Additionally, the results are overall similar across demographic groups and states, as well as using alternative weighting methodology. Finally, we discuss how our estimates compare to those from earlier papers that calculate estimates of bias in key CPS labor force statistics.

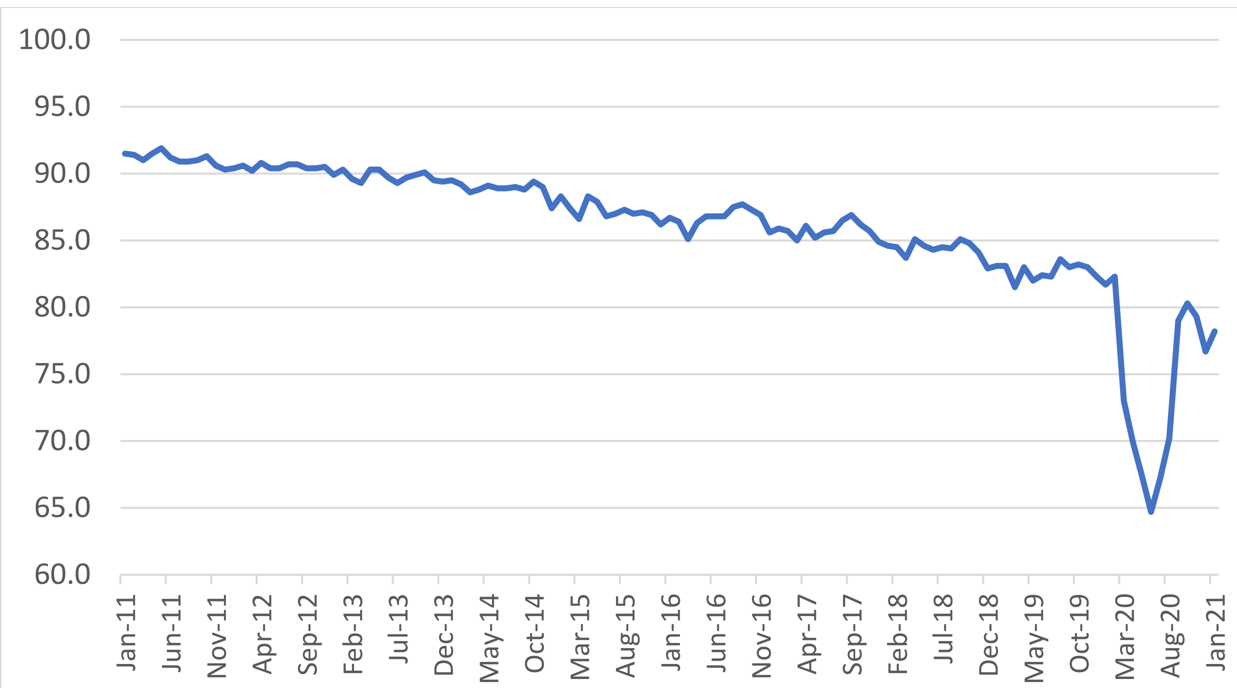
This paper is for research purposes only. No changes to production are being implemented at this time.

[*] Any opinions and conclusions expressed herein are those of the author and do not represent the views of the U.S. Census Bureau or the Bureau of Labor Statistics. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7521191: CBDRB-FY2021-CES010-018, CBDRB-FY2021-CES010-034, CBDRB-FY2022-CES019-009, CBDRB-FY2023-CES010-011).

1. Introduction

Declining unit response rates in surveys are a major concern for both data providers and users of survey data. In addition to surveys from academic institutions and the private sector, a variety of surveys administered by statistical agencies are also experiencing this trend (Czajka and Beyler 2016). This includes the Current Population Survey (CPS), the source of the official monthly estimates of the U.S. unemployment rate and other labor market indicators. During the 2010s, the response rate for the CPS declined from 91.5% in January 2011 to 81.3% in December 2019 (see Figure 1).[2] In addition to this longer run pattern, the COVID-19 Pandemic had a unique, acute effect on data collection when in-person interviewing was suspended for Census Bureau surveys on March 20, 2020. This shock resulted in CPS response rates dropping below 70% from April through July 2020.

Figure 1: CPS Response Rates



Source: Bureau of Labor Statistics

These declining response *rates* may increase the potential for nonresponse *bias* in the CPS unemployment rate and other key labor market statistics. Note that it is not necessarily true that

[2] This official response rate corresponds to American Association of Public Opinion Researchers (2016) RR6 definition.

nonresponse bias is increasing, as nonresponse rates are only weakly associated with nonresponse bias in surveys (Groves and Peytcheva 2008). However, several prior studies explore differences in response rates within the CPS sample, which may lead to questions about its representativeness. For example, Krueger, Mas, and Niu (2017) examine how the unemployment rate differs by the month a household is in the CPS sample. In the CPS, households are interviewed for four consecutive months, followed by an eight-month break in interviewing before another four-month interview period. Even though each month should be a representative sample on its own, the estimated unemployment rate for a household is higher in the first month the household is interviewed than other months. Additionally, Heffetz and Reeves (2019) find that respondents who are contacted with fewer attempts have higher unemployment rates and lower labor force participation rates than respondents who are more difficult to contact. If nonrespondents are similar to these harder-to-contact respondent households, then this also suggests a bias in the unemployment rate.

Nevertheless, Krueger, Mas, and Niu (2017) and Heffetz and Reeves (2019) are limited by the fact that they do not have information on nonrespondent households. Hence, it is unclear whether differences between respondents by month in sample or number of contact attempts can be used to estimate the characteristics of nonrespondents. When sampling frame or administrative data is available for nonrespondents, a more precise method to measure nonresponse bias is possible, such as with Lin and Shaffer (1995), who look at bias in reported child support payments, comparing them directly to individual-level data from Wisconsin Court Records. For unemployment, somewhat comparable administrative data exists for European countries, and Meraner et al. (2016) use these data to study nonresponse bias in labor force statistics in Austria. Unfortunately, no suitable administrative data for unemployment exists in the United States–nonemployment is observable in administrative wage records, but job search is not captured systematically by any government agency.

Even if a direct item-level comparison between survey and administrative data is unavailable for unemployment, there may be ways of using administrative data that are highly correlated with unemployment to construct a better measure of bias, and in turn, a more accurate estimate of the unemployment rate. Rothbaum and Bee (2021) match income data from the Internal Return Service (IRS) to respondents and nonrespondents in the 2020 CPS Annual Social and Economic Supplement (CPS ASEC). They use these data in an entropy-balance weighting (EBW) algorithm, as described in Hainmueller (2012a), to evaluate bias in income statistics for the CPS ASEC, finding an upward bias of 2.8 percent in median household income. In this approach, while there are conceptual differences between survey income and income data reported to the IRS, if these differences are inconsequential across respondents and nonrespondents, then comparing the results from the modified weighting approach to the official estimates should provide a reasonable approximation of nonresponse bias.

In this paper, we apply an approach like that of Rothbaum and Bee (2021) to the basic monthly CPS. We match income data from Internal Return Service (IRS) and state agencies to the sample at the address level, giving us annual and quarterly measures of employment and earnings. We also incorporate demographic and program benefit data from the Social Security Administration (SSA), demographic data from the decennial census, industry data from the Census Business Register, and home-value data from a third-party housing data vendor. Our primary weighting algorithm takes the matched administrative data, partitions households into mutually exclusive categories via classification and regression trees (CART), and then uses these categories to modify the household nonresponse adjustment portion of the CPS weighting algorithm. We also use an entropy-balance weighting algorithm with the same administrative data as a robustness check to examine how sensitive our results are to modeling choices. Our data on quarterly fluctuations in earnings, number of job holdings, and industry allow us to obtain detailed employment information, even if there is some loss in precision from inferring monthly employment status with our annual and quarterly data (because our measure of job search is annual unemployment insurance receipt, we have less precise information on whether a short nonemployment spell involved job search activity). Note that to be used in the weighting adjustments, these administrative data should be highly correlated with both response propensity and the survey outcome of interest (Little and Vartivarian, 2005). Hence, the data can be valuable even if there are measurement issues with the administrative data or conceptual differences in variables with the CPS.

After running the new experimental weighting algorithm with CART, we compare estimates of the unemployment and labor force participation rates using these experimental weights and the original production weights. Before March 2020, the experimental estimates of the labor force participation rates are 0.2 percentage points higher on average, but there are no economically meaningful differences in the unemployment rate. In April and May 2020, the experimental estimates of the unemployment rate are 0.2 percentage points higher, with smaller increases after that. The labor force participation rate continues to be about 0.2 percentage points higher for most months. The standard errors of these labor force statistics are similar between the experimental and original weights. In addition, the changes in the unemployment rate and labor force participation rate are fairly consistent across sex and race. However, there are some differences across education, with more educated groups exhibiting a larger increase in their estimated labor force participation rate. Additionally, results vary across states, although there are no apparent geographic patterns. Finally, we evaluate the EBW model and find that the results are largely similar to our main specification's.

Our results provide an alternate perspective on the magnitude of nonresponse bias in the CPS. To be clear, evaluating potential nonresponse bias in the CPS requires approximations and assumptions, as there is no administrative data source that exactly captures the same employment concepts in the CPS.[3] In this paper, we use the difference between the new and original estimates as an approximation of nonresponse bias, as although there are slight conceptual differences, our administrative data is highly correlated with employment and labor force participation. Previous literature has estimated larger sizes of nonresponse bias in the CPS. However, these results are reconcilable. Heffetz and Reeves (2019, 2021) develop a method to approximate the bias by comparing the unemployment and labor force participation rate of CPS respondents who respond within one or two contact attempts versus three or more contact attempts. Using this technique, Heffetz and Reeves (2021) approximate that the April 2020 unemployment rate could be biased upward by as much as 1.5 percentage points. While the direction of the bias they find is the same as our estimate, our magnitude is much smaller. But although they find modest different in employment rates by number of contact attempts, in terms of the effect on nonresponse bias, this differences is mitigated by the high response rate of the CPS, which may ultimately dampen the difference in employment status of nonrespondents.

In addition, our results contrast with the rotation group bias as described by Krueger, Mas, and Niu (2017), which suggests that the unemployment rate should be biased downward. Examining how rotation group bias changes between the original and experimental weights, we find no significant difference in the magnitude of rotation group bias across the two sets of weights. This may suggest that rotation group bias is due more to measurement error than nonresponse bias, contra Krueger, Mas, and Niu (2017).

2. Overview of the Current Population Survey

The Current Population Survey is the key source of labor force statistics for the United States and has existed in some form since the 1940s. Respondent households in the sample are interviewed monthly. Each month contains a basic battery of demographic and labor force questions, with periodic

[3] For example, although states have weekly data on who received unemployment insurance, someone can be unemployed without receiving unemployment insurance.

supplements covering additional topic areas. To reduce sampling error in estimates of month-to-month changes, addresses are interviewed repeatedly for a limited duration. An address is in the CPS for four months, followed by an eight-month break, followed by another four-month interviewing period. Interviews for the first and fifth month are typically done in-person, with telephone interviews in other months. The current monthly sample size of the CPS is about 60,000 households.

2.1 Sampling and Weighting

Sampling in the Current Population Survey is a two-stage process. The first step is determining which geographies are selected for the sample. In this step, the U.S. gets divided into primary sampling units (PSUs) which are counties or groups of counties. For large population areas, like Metropolitan Statistical areas, these primary sampling units are always in-sample, denoted as self-representing (SR). For smaller areas, the remaining primary sampling units are stratified by state, and one primary sampling unit is selected from each stratum, denoted as non-self-representing (NSR) primary sampling units. In the second step, households within a PSU are sampled for the CPS. To reduce field costs, households are clustered at the block level.

The Current Population Survey weighting process consists of three major steps: a unit nonresponse adjustment, a ratio adjustment to account for first-stage primary sampling unit selection, and a series of raking steps across national and state breakdowns by various demographic cells.

Starting with the base weights, households are assigned to nonresponse clusters based on whether they fall within a metropolitan statistical area. Within metropolitan nonresponse clusters, these are broken down into principal city and non-principal city cells. The metropolitan cells can be collapsed together if the size of the nonresponse factor is too large, if the cell has too few households, or if either cell has no respondents.

After the nonresponse adjustment, the first-stage factor is applied to the individuals within each household. This factor adjusts for Black alone/not Black alone population counts in the NSR primary sampling units. Finally, a series of benchmarking steps is performed using monthly population estimates, first for age/race/ethnicity/sex cells at the national level, then state level race/sex/age groups, and then a sequence of raking steps is conducted based on state, ethnicity, and race cells. These produce the second-stage weights for the civilian noninstitutionalized population. For more details on the weighting procedures, see U.S. Census Bureau (2019).

2.2 Interviewing During COVID-19 Pandemic

During the COVID-19 pandemic, there were interviewing restrictions on all surveys conducted by the U.S. Census Bureau. In-person interviewing was suspended on March 20, 2020, near the end of the interviewing week for the March 2020 CPS. Starting in July 2020, limited in-person interviewing resumed in some parts of the country, and call-centers resumed operations in July 2020 in a limited capacity. Throughout the rest of 2020 and most of 2021, in-person interviewing was allowed in areas based on local COVID-19 case counts and related metrics. Further details on Census interviewing during 2020 and 2021 can be found in U.S. Bureau of Labor Statistics (2021) and U.S. Census Bureau (2020).

These restrictions impacted response rates and nonresponse bias in the CPS. First, the absence of in-person interviewing prevented initial in-person interviews for the first month a household was in-sample, as well as in-person follow-up for some cases in other months. This reduced the ability to convert refusals, contributing to lower response rates.

In addition to refusals, there were also implications for surveyors' ability to contact households. Before COVID-19, when households were interviewed over the phone, the interviewer used the respondent-provided phone number from the first interview. For households that had already been interviewed prior to March 2020, Census staff thus had high-quality phone numbers for contacting respondents. However, for households entering the sample after March 2020, interviewers had to rely instead on the Census Bureau Contact Frame. The Contact Frame consists mainly of phone numbers from a commercial provider, with additional information from sources such as prior American Community Survey (ACS) responses and Supplemental Nutrition Assistance Program data for certain households.[4] These phone numbers were of varying quality, and some households did not have a valid number matched to their address. Eggleston (2021) presents evidence that lower income households were less likely to have a valid phone number in the contact frame, implying that whether Census had a usable phone number for an address may be non-random.

3. Administrative Data

Our primary administrative data consist of annual tax data from the Internal Revenue Service (IRS). First, we use IRS Form 1040 individual income returns. The 1040 returns include income from the prior year and the identities of the first four children or other family members claimed as dependents. Next, we

[4] The Contact Frame now also contains phone numbers provided in the 2020 Census, but these were not incorporated in the contract frame during the time period analyzed in this paper.

use IRS information returns, which include W-2s from employers, 1099 forms such as the Form 1099-INT, and Form 1098. These forms give us additional information on a variety of sources of income, as well as broader economic and financial activity. Receipt of a 1098 from a mortgage lender indicates whether the household has a mortgage, a proxy for home ownership. Filing status from a 1040 return also gives us a proxy for marital status. Our matching and data construction procedures resemble those of Eggleston and Westra (2020), who examine administrative data-based weights for the Survey of Income and Program Participation (SIPP). Our work for the CPS uses some additional data sources, such as program benefit data from the Social Security Administration (SSA). The tax year we use matches the CPS year in question. For example, we use the 2020 IRS data to adjust the weights for the April 2020 CPS, and this data would reflect the income a household received during the entirety of 2020, not just the CPS month in question. These data come from tax returns that would have been filed in 2021, after the CPS estimates are released. In Appendix A.1, we show estimates using only administrative data that would have were available at the time the CPS data is prepared for release. Overall, the estimates are very similar.

To match survey and administrative records at the address level, we utilize the linking identifier in the Master Address File (MAF), the Census Bureau's frame file of all known living quarters and certain nonresidential addresses in the United States. Since 2014, the MAF has been used as the sampling frame for the CPS, so all CPS observations have this linking identifier, called the MAFID. The IRS data are linked to the MAF using a probabilistic linking algorithm. Looking at 1040 data, Bee, Gathright, and Meyer (2015) find that about 90 percent of Form 1040 records match to a MAFID, with unmatched tax records tending to be in the extreme upper and lower ends of the income distribution. There are two possible reasons for why a household does not match to IRS data. It could be that this household did not file their taxes or receive any other tax form (among those provided to the Census Bureau) from an employer or other party. Or it could also be that they did have one of these tax forms but a problem with the address listed on their tax form resulted in no MAFID being assigned. While this non-linkage makes the reweighting less efficient, it remains consistent if the relationship of non-linkage to the data is independent of survey response.

3.1. Construction of Household Roster

After linking several administrative datasets to the CPS by address (MAFID), our second step is to use the administrative data to construct a household roster for each address that we then use as a basis for matching additional person-level datasets. Our primary sources for constructing a household roster are the IRS 1040 returns and the IRS information returns (e.g., Forms 1098, 1099, and W-2). Importantly, the dependent data in the 1040s gives us information about children in the household. Using the information returns in addition to the 1040s is useful for picking up individuals who do not have any filing requirements. For example, retired individuals who only receive Old-Age, Survivors, and Disability Insurance (OASDI) benefits typically do not have any taxable income. These individuals then might not file an IRS 1040 tax return, but they would receive an SSA-1099, allowing us to add them to household rosters.

However, some people in the United States do not appear in any IRS data available to the Census Bureau. To capture some of these individuals, we utilize two additional sources when constructing the household rosters. First, we use location information from Census Bureau's Master Address File Auxiliary Reference File (MAFARF). The MAFARF contains residency information from IRS data and other administrative data sources, such as the United States Department of Housing and Urban Development (HUD). Second, we use data on Supplemental Security Income (SSI) recipients from SSA's Supplemental Security Record (SSR). Since SSI is not taxable, individuals whose sole source of income is SSI would not appear in IRS data. Together, these two data sources improve our ability to identify people not present in the IRS data.

Nevertheless, the administrative record-based household roster we construct may still miss some individuals who would be residents of the address at the time of the CPS interview. Additionally, our rosters are all based on annual data. With movers, our administrative data could correspond to an entirely different household at the same address. However, our reweighting exercise does not require us to replicate exactly the household roster that appears (or, for nonrespondents, would appear) in the CPS. Instead, we merely need to construct a *proxy* of who resides in each household. The more closely correlated these proxies are to dimensions of the data that are correlated with response, the more efficient the weighting becomes. For example, consider the children of divorced parents. A custody agreement may result in a difference between which parent claims the child on the 1040 in a year and where the child is deemed to reside in accordance with CPS residency rules. However, if these discrepancies are similar between respondent and nonrespondent households, the reweighting should remain consistent. To the extent that these discrepancies are relatively rare, they result only in small reductions in how well our administrative data variables are correlated with the CPS survey data, and thus the efficiency of the reweighting process.

3.3. Additional Data Linkage

Given the household roster, we link several additional datasets to learn more about these households, as depicted graphically in Figure 2. We use demographic data from the SSA's Numident file and the 2010 Decennial Census. Both the Numident and 2010 Census data contain information on an individual's age and race. If race data are available from both sources for a person, we use the decennial census race. The 2010 Census also provides information on Hispanic origin, while the Numident data contain information on citizenship and foreign-born status. We also link quarterly earnings data from the Longitudinal Employer-Household Dynamics (LEHD) program, which is itself is derived from state unemployment insurance earnings data and the Quarterly Census of Employment and Wages. The main benefit of these data is that they provide earnings at the quarterly level, while the IRS data is annual.

These datasets are linked to the IRS-based household rosters at the individual level using the Census Bureau's Person Identification Validation System (PVS), as described by Wagner and Layne (2014). This procedure matches both survey and administrative data to a master reference file. Individuals who are matched are given an identifier called a Protected Identification Key (PIK), which acts as an anonymized Social Security number that can be used to link administrative datasets and surveys. For the 2010 Census, about 90 percent of individuals were matched to a PIK (Wagner and Layne, 2014).[5] Bond et al. (2014) argue that inability to match to a PIK is nonrandom. They use 2009 ACS data to document that young children, minorities, immigrants, recent movers, low-income individuals, and non-employed individuals are less likely to be matched to a PIK. Our reweighting algorithm accounts for this nonrandom ability to observe characteristics in administrative records linked to the household roster by PIK.

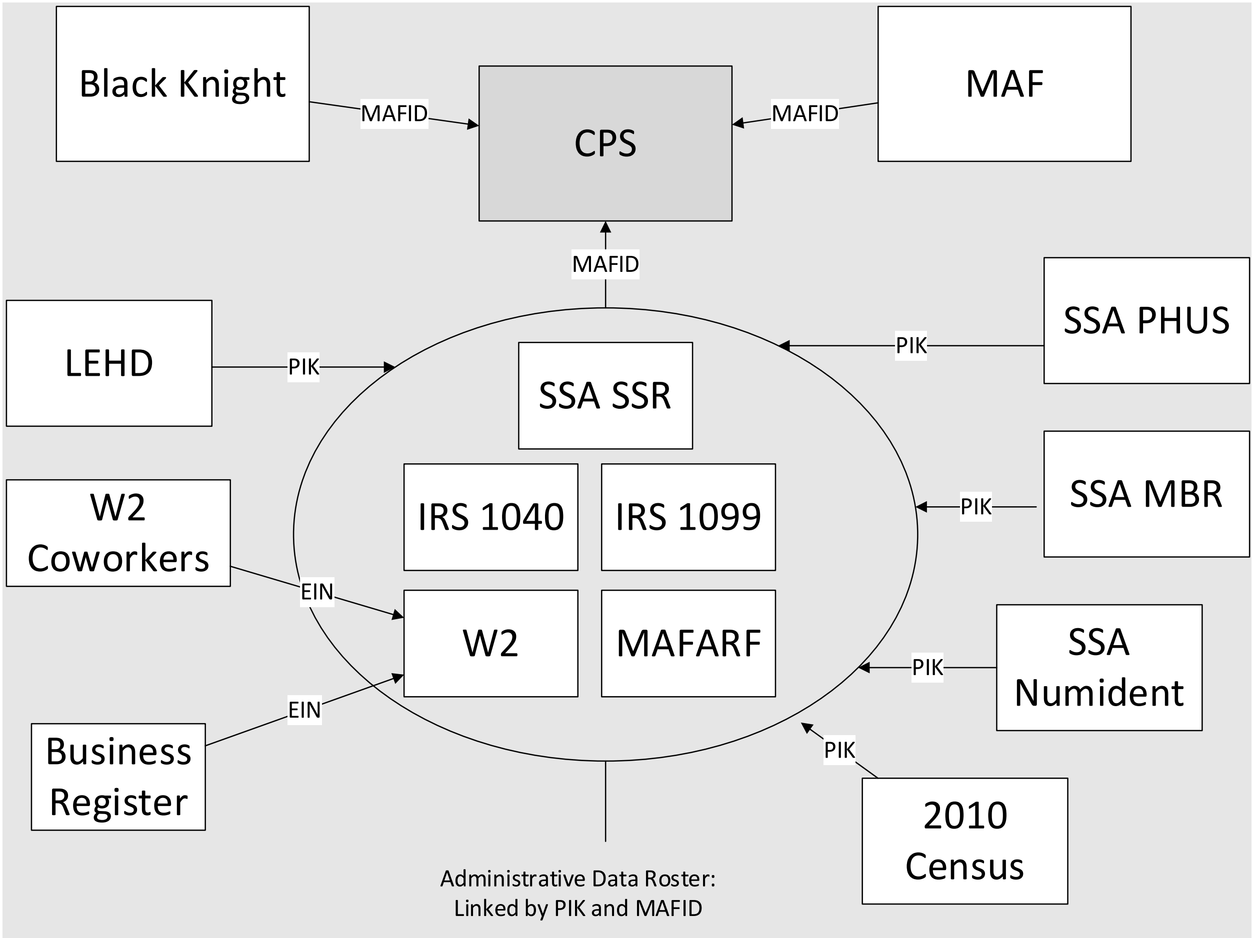
Once the Numident and 2010 Census data are matched at the individual level, the data are then aggregated to the household level to create address-level measures for comparing respondent and nonrespondent households. For example, we create an indicator for the presence of a household member over age 60 by taking the list of people on the tax forms matched to an address and matching these people to the Numident file to get the year of birth for all household members. Additionally, we match OASDI program benefit data for SSA's Master Beneficiary Record (MBR) and the Payment History Update System (PHUS) at the person level using PIK.

[5] The 2010 Census is also linked to the MAF. Because many people have changed their residential location since 2010, however, we link decennial data at the person level instead, to make sure we are capturing the characteristics of the current household.

For obtaining information about employer characteristics, we use the Employer Identification Numbers (EIN) on the W-2s matched to the household. First, we leverage the W-2s of a household member's coworkers to obtain a proxy of how many people work at their employer in case there are unemployment shocks in a month that are correlated with firm size. Second, we use the Census Bureau's Business Register to obtain information on the industry each household member works in. Given that the employment of people in the hospitality and food service industry was particularly affected by COVID-19 restrictions, this industry information may be especially valuable for economic outcomes in 2020. Note that this linkage by EIN is imperfect for many businesses, as EIN is neither a firm nor an establishment identifier for multi-establishment firms. In other words, the number of people who share the same EIN may not always equal the number of people that work at a person's place of business. However, since our goal is to construct proxies for employer characteristics rather than to construct precise establishment-level statistics, and we do not expect the validity of these proxies to vary by survey response, we believe the EIN linkage is adequate as an input for our weighting algorithm.

Finally, we link commercial housing data from Black Knight, a third-party vendor, and structure-type information from the MAF. From the Black Knight data, we use information on the home's assessed value as well as whether it is owner-occupied. The home's value and whether the residence is a single-family home or in a multi-unit structure provide information on the socioeconomic status of households, particularly for households that we are unable to match to any other administrative data.

Figure 2: Data-Matching Diagram



3.3. Linkage Rates

Table 1: Linkage Rates to Administrative Data

CPS Year and Tax Year 2019	Respondent	Nonrespondent
Match to 1040 Tax Return	81.29	79.11
Match to Any 1040, 1099, SSR, or MAF-ARF Record	87.34	84.01

Table presents address-level match rates to the 2019 CPS. Source: 2019 CPS, IRS 1040, IRS 1099, SSR, and MAF-ARF

Households may fail to link if there are either actually missing from the administrative data or were present in the administrative data but their administrative record could not be assigned a MAFID. Table 1 presents linkage rates for 2019. Among respondents, 81.3% of households were matched to an IRS 1040 tax return, and 87.3% were matched to any of our main administrative datasets (1040, 1099, SSR, or MAF-ARF). Rates for nonrespondents are similar but are a 2-3 percentage points lower. Overall, more than 10% of CPS households are missing from our primary administrative data. For these households, we can still observe a few other pieces of information, such as the building structure type from the sample frame and a home's assessed value from the third-party housing data.

For person-level linkages conditional on the CPS sampled addressed being matched, PIK assignment is very high. In tax year 2018, 99.45% of all 1040 person-level records were successfully assigned a PIK. And the Numident has a 100% PIK assignment rate, as it is the master file of Social Security numbers. This means that we have age, sex, citizenship, and foreign-born information on almost all individuals present in our administrative data records. Only race and Hispanic origin information are less complete because the PIK assignment rate for their source, the 2010 Census, is around 90%.

4. Incorporating Administrative Data in Weighting

Matching CPS households to administrative data at the address level provides us with additional variables to consider when creating the nonresponse adjustment cells. In the original CPS nonresponse adjustment, there are essentially just two variables used, the state and an urban/rural indicator, so these variables can be cross-tabulated to create mutually exclusive categories. However, the addition of the administrative data creates many new variables, so simple cross-tabulation will be infeasible. Thus, the administrative data needs to be collapsed in some way to create a manageable number of adjustment cells.

When creating these cells, it is important that the variables used are correlated with both response propensities and the survey outcomes of interest. Little and Vartivarian (2005) show that if this is the case, then weighting with such a variable will reduce both bias and standard errors. However, if a weighting variable is only correlated with response but not the survey outcomes, then adding this variable will have no effect on bias while increasing standard errors. Because of the importance of using variables that are correlated with employment outcomes, we use a classification and regression tree (CART) model to help collapse the administrative data. CART is a nonparametric model used to predict the value of a dependent variable from several explanatory variables. Because it is nonparametric, it easily incorporates interactions, which may be important for predicted employment status (e.g., having no earnings in a year interacted with the presence of children in a household might be highly predictive of being out of the labor force). One particular benefit of CART is that it utilizes a pruning procedure to prevent overfitting. In addition to providing statistical benefits, CART also provides practical benefits given its integration with the HPSPLIT procedure in SAS, the language in which the CPS data cleaning programs are written. This ease of incorporation may make modifications to the official CPS weighting procedures more feasible.

To operationalize CART for CPS weighting, we create a household-level recode of the employment statuses of household members. Specifically, looking at just the people in the household who are over 15 and in-universe for the main employment questions, we create a categorical variable that equals one if anyone in the household is unemployed; two if no one is unemployed and at least one person is disabled; three if no one is unemployed or disabled but at least one person is retired; four if no one is unemployed, retired, or disabled, but at least one person is not in the labor force for another reason; and five if every adult in the household is employed.

We estimate this model on respondents. We then use the resulting decision rules for the CART leaves to assign nonrespondents to the appropriate noninterview cell. We impose the restriction that each cell has at least 50 respondent households to ensure the stability of the adjustments. For the CPS weighting procedures, we replace the geography-based noninterview adjustment with the cells derived from this CART model but leave other parts of the CPS weighting algorithm intact, such as the calibration of weights to population estimates.

The CART model we have specified has several features beneficial for weighting the CPS data. First, by using the aforementioned employment status dependent variable in the CART model, the CART procedure gives more weight to the administrative data that is most correlated with response outcomes. If, for example, the proxies for firm sizes derived from coworker W-2s have a low correlation with employment outcomes (after controlling for other factors), then these variables will have low impacts on the CART cell definitions. Second, to handle non-linkages, we use the match indicators as explanatory variables in the CART model. For example, Table 1 in the previous section shows that nonrespondent households were less likely to be matched to any of our administrative data. By using linkage as a predictive variable, we can adjust for the lower response rates of the non-matched households. Thus, non-linkages with administrative data are not a problem for the CART model per se. The main issue is that for any noninterview adjustment cell of non-matched households, the missing conditionally at random assumption implicit in the weighting process is more questionable for these households than for other cells with more detailed administrative data. Third, the CART model is run separately for every month. If the relationship between the survey and administrative data changes over time, then the CART model can account for that. For example, our proxy for a household member working in the restaurant industry is more predictive of unemployment in 2020 than in the 2010s, and the CART model can account for this. These monthly updates can also account for changes in the tax code, like with unemployment insurance rules for self-employed individuals during the COVID-19 pandemic, or tax filing behaviors, as with the increase in people with no income filing to receive COVID-related relief payments.

Table 2: CART Model Stability

Rank	Variable	Relative Importance	Percent of Time This Variable is in Top 10 of Selected Variables
	May 2019		
1	Anyone Over 60 Receiving SSA OASDI Benefits	1	1
2	Household Structure Type	0.4966	1
3	Receipt of Any OASDI Disability Benefits	0.3997	0.5556
4	Someone Between Ages of 40 and 60 in Household Who Isn't Working (No W-2 or Schedule SE)	0.3554	1
5	Someone Over 60 in Household Not Working (No W-2 or Schedule SE)	0.3446	1
6	Retirement Income	0.2244	0.5741
7	Number of W-2s for Person in Household with the Most Jobs	0.2234	0.8889
8	Someone between 10 and 17 in Household	0.2126	0.6481
9	Receipt of Any SSI Disability Benefits	0.2064	0.2222
10	Any Person in Household with the Same Employer as Last Year (W-2 with same EIN)	0.2024	0.6481
	May 2020		
1	Anyone Over 60 Receiving SSA OASDI Benefits	1	1
2	Household Structure Type	0.4921	1
3	Receipt of Any OASDI Disability Benefits	0.3718	0.5556
4	Arc Percent Change in Earnings Between Q1 and Q2, the Smallest Amount for Anyone in the Household	0.3632	0.07407
5	Number of W-2s for Person in Household with the Most Jobs	0.3317	0.8889
6	Someone Between Ages of 40 and 60 in Household Who Isn't Working (No W-2 or Schedule SE)	0.3205	1
7	Someone Over 60 in Household Not Working (No W-2 or Schedule SE)	0.2704	1
8	Any Person in Household with the Same Employer as Last Year (W-2 with same EIN)	0.2632	0.6481
9	Receipt of Any SSI Disability Benefits	0.2298	0.2222
10	Someone Between Ages of 26 and 40 in Household Who Isn't Working (No W-2 or Schedule SE)	0.1716	0.8704

Table present measures of variable important for the CART model used to create the administrative data-based weighting cells. The table gives the CART SSE measure of relative importance, as well as the percent of times this variable appears in the top 10 list for the other months. Source: 2019 and 2020 CPS matched to administrative data.

While the CART model gives flexibility, the chosen variables in the model ended up being stable. Table 2 describes the top 10 most important variables selected in the CART Model for the May 2019 and May 2020 CPS, showing the CART SSE measure of relative importance and how often this variable is one of the 10 most important in other months. The most important variable is whether there is anyone over 60 receiving SSA benefits in the household, followed by the household structure type. This variable we have constructed from other demographic information in our data describes how many adults and children are in the households. Receipt of OASDI Disability Benefits is only in the top 10 variables for importance 55% percent of the time purely because we do not have this SSA data in 2017 and 2018. In May 2020, the change in earning from the LEHD is the 4th most important variable, but it only has substantial predictive power in that month.

4.1. Explanatory Power of Administrative Data Variables

In our administrative data-based weighting method, we can obtain valuable data on the economic characteristics of nonrespondent households, but because our administrative data do not contain direct measures of unemployment, we must consider its application carefully. The administrative data do reasonably well at identifying whether someone worked at all in the prior year. For example, we can see whether someone received any retirement or social security income, or whether they had any employment income reported through a W2 or filed schedule SE for their 1040.[6] Thus, for people out of the labor force for at least a year, the administrative data likely provides a good indication of this status.

These data are less likely to allow us to detect shorter spells of nonemployment. Our administrative data contain information on possible employment volatility, such as having several jobs the prior year, having lower income, or working in the accommodation and food services industry. For the COVID-19 pandemic, the state earnings data allow us to see individuals who had a large drop in earnings between the 1st and 2nd quarters of 2020, which would likely indicate individuals whose employment was affected by the pandemic and may have been unemployed or out of the labor force in the 2nd quarter of 2020.

[6] Note that we do not use receipt of a 1099-MISC in this definition. Census only receives data on whether someone received a 1099-MISC, but not which box was filled on the form. In addition, IRS released instructions in 2020 that form 1099-NEC should be used now to report self-employment income. Census does not receive data on 1099-NEC receipt. Individuals are required to file taxes if they have more than $400 in self-employment income, so presence of a schedule SE should capture most of any self-employment income that would have been reported on a 1099-MIS or 1099-NEC.

However, these measures are only at the annual or quarterly level, so their timing might not line up with the CPS week in question.

Additionally, our data on receipt of unemployment insurance is lacking, which further contributes to our difficulty in separating unemployment from being out of the labor force for a short period of time. In the IRS 1040 data the U.S. Census Bureau receives from IRS, only a few income fields are given. In particular, Census does not receive the unemployment insurance field from the 1040s. Census does receive information on whether someone received a 1099-G, but we do not know whether this was for unemployment insurance, a state refund for itemizers, or another reason. In the CART model, we interact the 1099-G indicator with whether the person itemizes their tax returns to improve the predictive power of the variable, but this still only gives some information on whether someone received unemployment insurance at all during the year without data on exact timing. Additionally, there are many differences between the requirements to receive UI and the CPS definition of being unemployed, like people who quit their jobs being ineligible for UI and employed individuals with reduced hours who were eligible for UI during the pandemic. Although the 1099-G indicator is only correlated with employment status rather than being a perfect measure, it should still improve nonresponse bias despite the differences.

Table 3-Relationship Between Survey and Administrative Measures of Employment

	Percent Households with an Adult (25+) Employed, conditional on administrative data value	
Someone with a W2 or 1040-SE?	2019	April-August 2020
No	34.85	33.01
Yes	86.36	85.03

Source: Basic Monthly CPS matched to administrative data. Table presents household-level means from the CPS survey data conditional on a value from the administrative data. CPS months used are all of 2019 and April through August of 2020.

Table 4-Relationship Between Survey and Administrative Measures of Not Working

	Percent Households with an Adult (25+) Not in Labor Force	
Someone *without* a W2 or 1040-SE?	2019	April-August 2020
No	28.98	29.02
Yes	77.07	77.36

Source: Basic Monthly CPS matched to administrative data. Table presents household-level means from the CPS survey data conditional on a value from the administrative data. CPS months used are all of 2019 and April through August of 2020.

Table 5-Relationship Between Survey and Administrative Measures of Not Working

	Percent Households with an Adult (25+) Retired or Disabled	
Any OASDI or SSI SSA Benefits?	2019	April-August 2020
No	17.38	17.12
Yes	76.15	76.28

Source: Basic Monthly CPS matched to administrative data. Table presents household-level means from the CPS survey data conditional on a value from the administrative data. CPS months used are all of 2019 and April through August of 2020.

Given this context of the strengths and weaknesses of the administrative data, we examine the misclassification rate of the household-level employment variable we use in the CART model. First, Tables 3-5 examine the predictive power of the administrative data which give annual measures of working status. Table 5 shows the relationship between the household having a W2 (indicator of working for an employer) or filing a 1040 Schedule SE (indicator of self-employment) and whether anyone in the household is reported as being employed in the CPS.[7] The statistics pool all the CPS months together, and no person-level linking across months is performed. Thus, one person will appear

[7] We use an age range of 25+ (which matches CPS educational attainment tables) to avoid complications with younger adults who may still be in college.

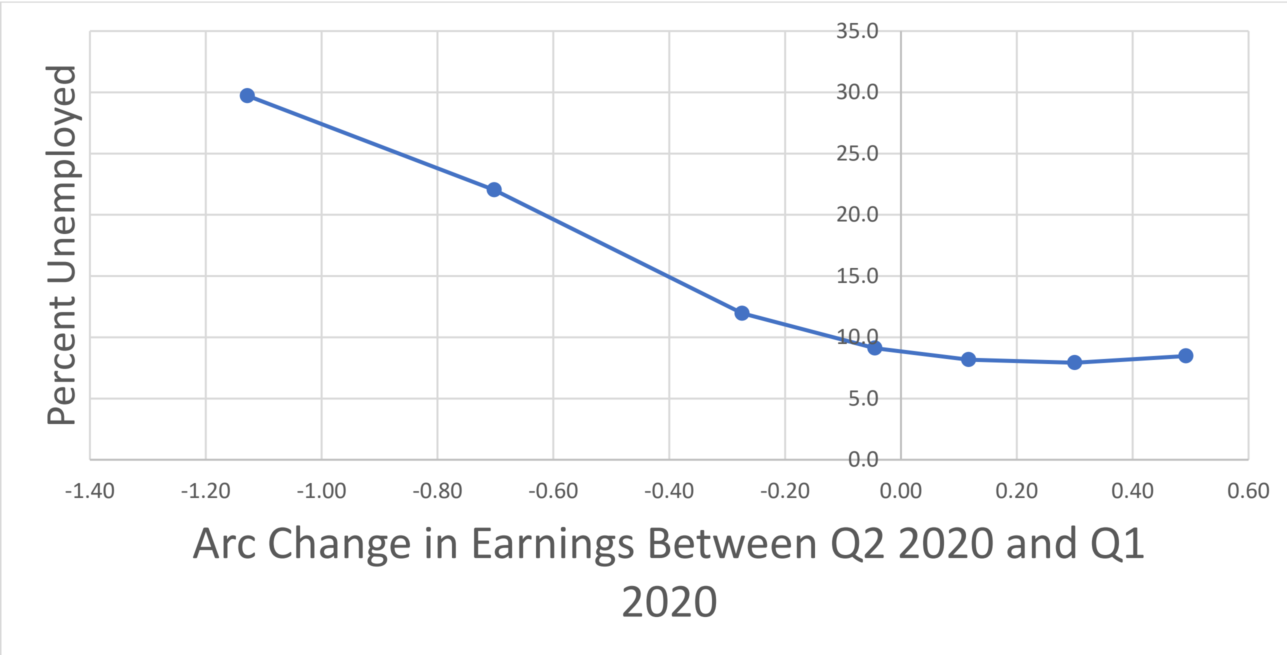
multiple times in the analysis sample for this table and may have different employment statuses in different months.   The administrative data are an annual measure corresponding to the tax year in the column header that does not vary by month. The number in the lower left shows that of households where someone have a W2 or Schedule-SE, 86.36% of these households report having an employed adult in the January – December 2019 CPS. This rate is only 34.85% for households without a W2 or 1040 Schedule-SE. Overall, this shows the administrative earnings data is very informative of employment in the CPS.

Despite the close links between administrative earnings data and CPS employment, there are some differences. An obvious explanation is the conceptual differences between these measures of employment. For example, self-employed individuals with less than $400 in earnings are not required to file taxes, so they may also be missing from IRS data. Another contributor may be differences in household roster across survey and administrative data. There could also be temporal components to the discrepancy: the CPS data measure employment in a particular week of a particular month, while the administrative data indicate whether the person was employed at any point in the year. Additionally, differences between the administrative data-based roster and the CPS roster (particularly with movers) described in Section 3.1 could also cause some of the differences in this table.

Table 4 also shows a similar trend looking at people not in the labor force and whether there is any adult over 25 in the household with a W2 or 1040 Schedule SE, and Table 5 examines the relationship between being retired or disabled in the CPS and receiving social security benefits. The relationship between the survey data on labor force participation and the administrative data is reasonably strong, suggesting that these data can be used to detect bias in the labor force participation rate.

Figure 3 Percent Households with an Adult (25+) Unemployed & Earnings Changes

April-June 2020



Source: April, May, and June 2020 Basic Monthly CPS matched to administrative data. Figure shows how a household's change in administrative earnings from the LEHD between the 1st and 2nd Quarter of 2020 is related to whether anyone in the household is unemployed. Sample consists only of households with positive earnings in the 1st and 2nd quarter. Estimates are from Kernel-weighted local polynomial smoothing with a Gaussian Kernel. Grid points are the 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles, which are connected linearly in Figure 3.

Next, we examine how well our administrative data capture unemployment. Our strongest predictor comes from quarterly changes in earnings. Figure 3 shows how changes in earnings between the 1st and 2nd quarter of 2020 relate to whether someone in a household reported being unemployed in the April-June 2020 CPS. The 1st quarter represents earnings received between January 1st and March 31st, and the 2nd quarter represents earnings from April 1st to June 30th. Given the first stay-at-home order in the U.S. was California's on March 19th, 2020, most of the initial economic effects of the COVID-19 shutdowns would appears in 2nd quarter earnings, so these quarter-to-quarter changes should be informative for the time when in-person interviewing was completely suspended. Note that this proxy may not be as useful for future unemployment shocks if the timing does not line up as well with calendar quarters as well as it did for the COVID-19 pandemic.

On the horizontal axis is the arc percent change in quarterly earnings, which for earnings $E_{i,t,q}$ for person i in quarter q in year t is

$$\frac{(E_{i,2020,2} - E_{i,2020,1})}{abs(E_{i,2020,2} + E_{i,2020,1})/2}$$

Unlike percent changes, the arc percent change is the same in absolute value regardless of whether the 1st or 2nd quarter is taken as the initial value, and it is bounded between -2 and 2. In order to focus on households that remain in the labor force rather than exiting, we limit the sample for the graph to households with positive earnings in both the 1st and 2nd quarters of 2020.

The graph shows that the change in quarterly earnings is predictive of unemployment status. For households at the fifth percentile in change of earnings, about 30 percent have at least one household member who is unemployed. For households with only a small decrease in earnings or an increase, this rate is below 10 percent. These differences are not as large as in Tables 3-5 for differences in labor force participation by administrative data values. Because people without a job could be either unemployed or out of the labor force, a lower correlation here is not too surprising. Therefore, while these data are predictive of unemployment status and should be useful for detecting the direction of any bias in the unemployment rate, weighting adjustments which use these data may not perform as well in correcting bias in the unemployment rate as they do in the labor force participation rate.

5. Results

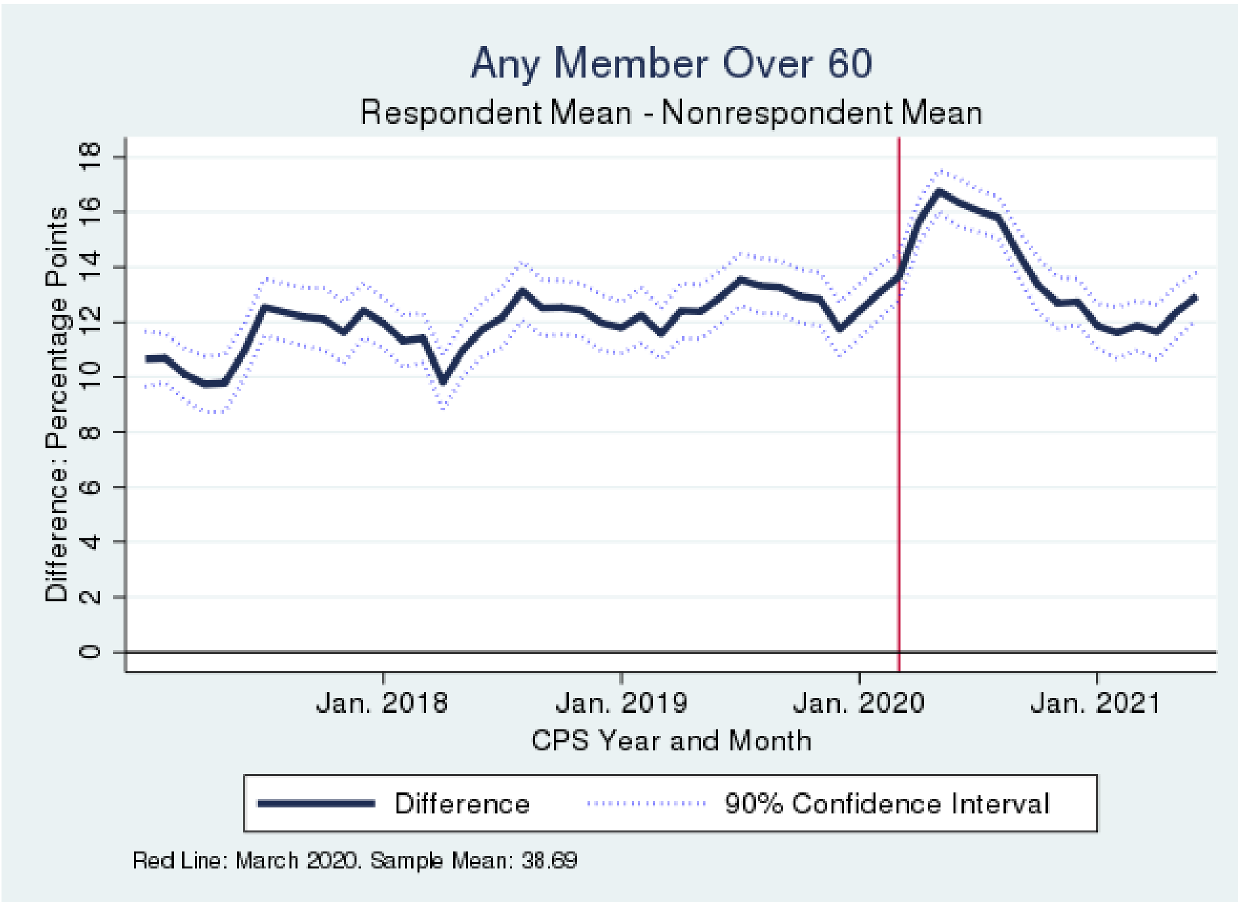
5.1. Change in Respondent Sample

Before showing how the alternative weighting methodology changes CPS estimates, we first present results comparing the characteristics of respondents and nonrespondents over time. In Figures 4-8, we calculate means of administrative data-based variables for respondents and nonrespondents in each CPS month and plot the differences over time.[8] Only base weights are used on these graphs, so in essence, these graphs indicate the magnitude of bias before any weighting adjustments. Because nonresponse bias can be decomposed into the product of the nonresponse rate and the average difference between respondents and nonrespondents (Groves and Peytcheva 2008), these graphs indicate the degree of

[8] Replicate weights are used to construct the standard errors of these differences. That is, the difference is calculated for each replicate, and we use this information to construct the standard errors. Given we have nonrespondents in this analyses, the replicate weights used here are the ones from the weighting algorithm right before the noninterview adjustment.

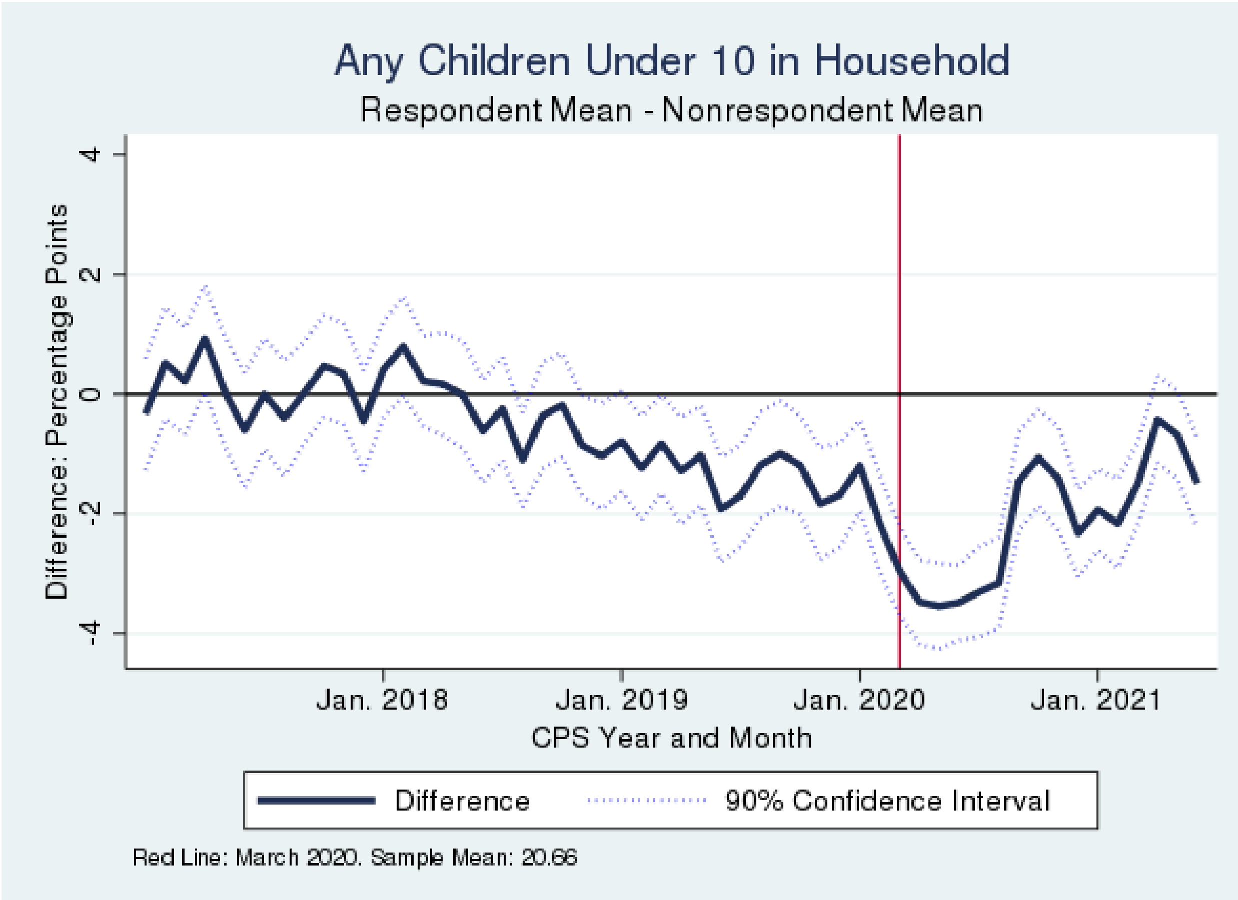
representativeness of the respondent sample holding aside any changes in response rates, which we know decreased in 2020.

Figure 4: Nonresponse Comparison for Any Member over 60



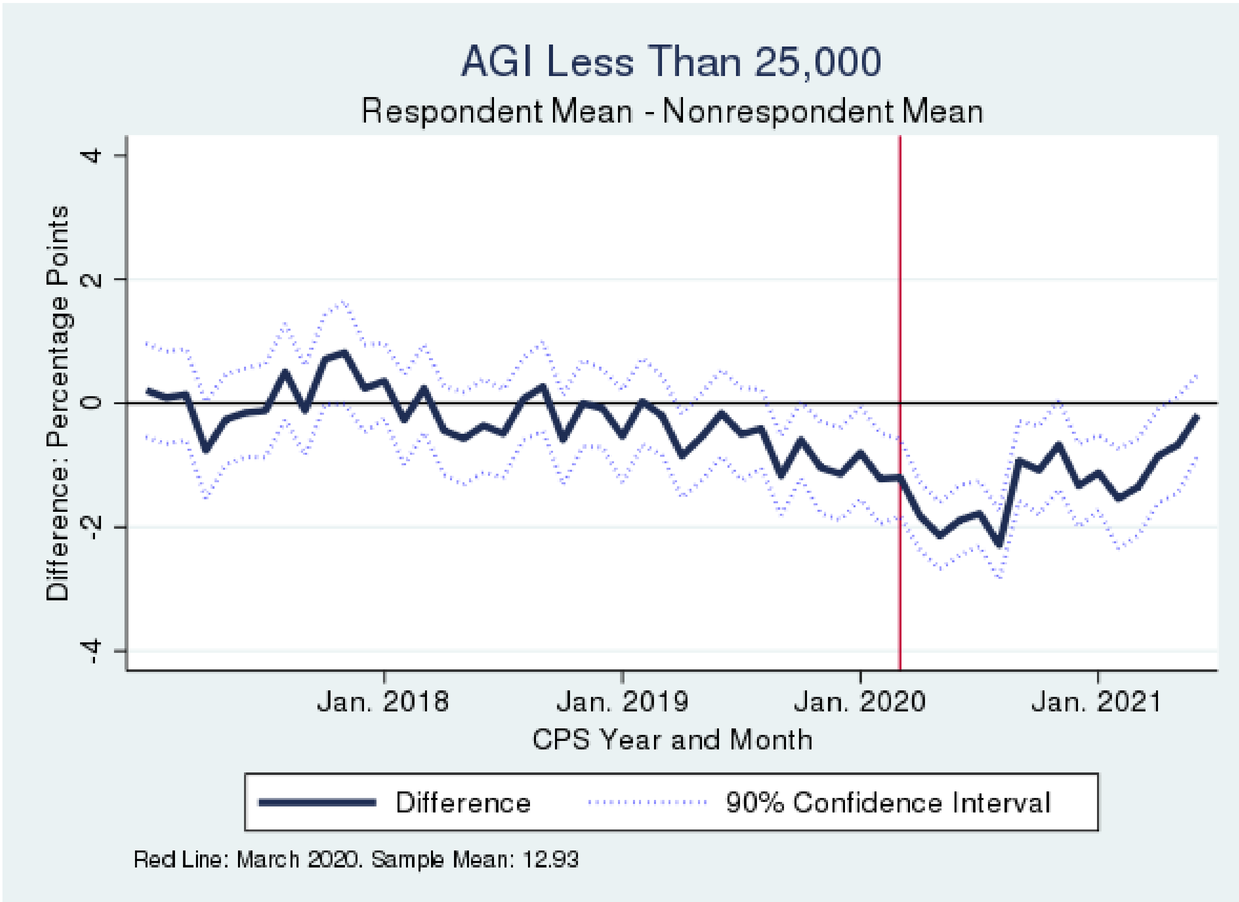
Source: Basic Monthly CPS matched to administrative data. Figure shows for each CPS month, the average for respondents minus the average for nonrespondents. CPS months used are January 2017 to June 2021.

Figure 5: Nonresponse Comparison for Any Children Under 10



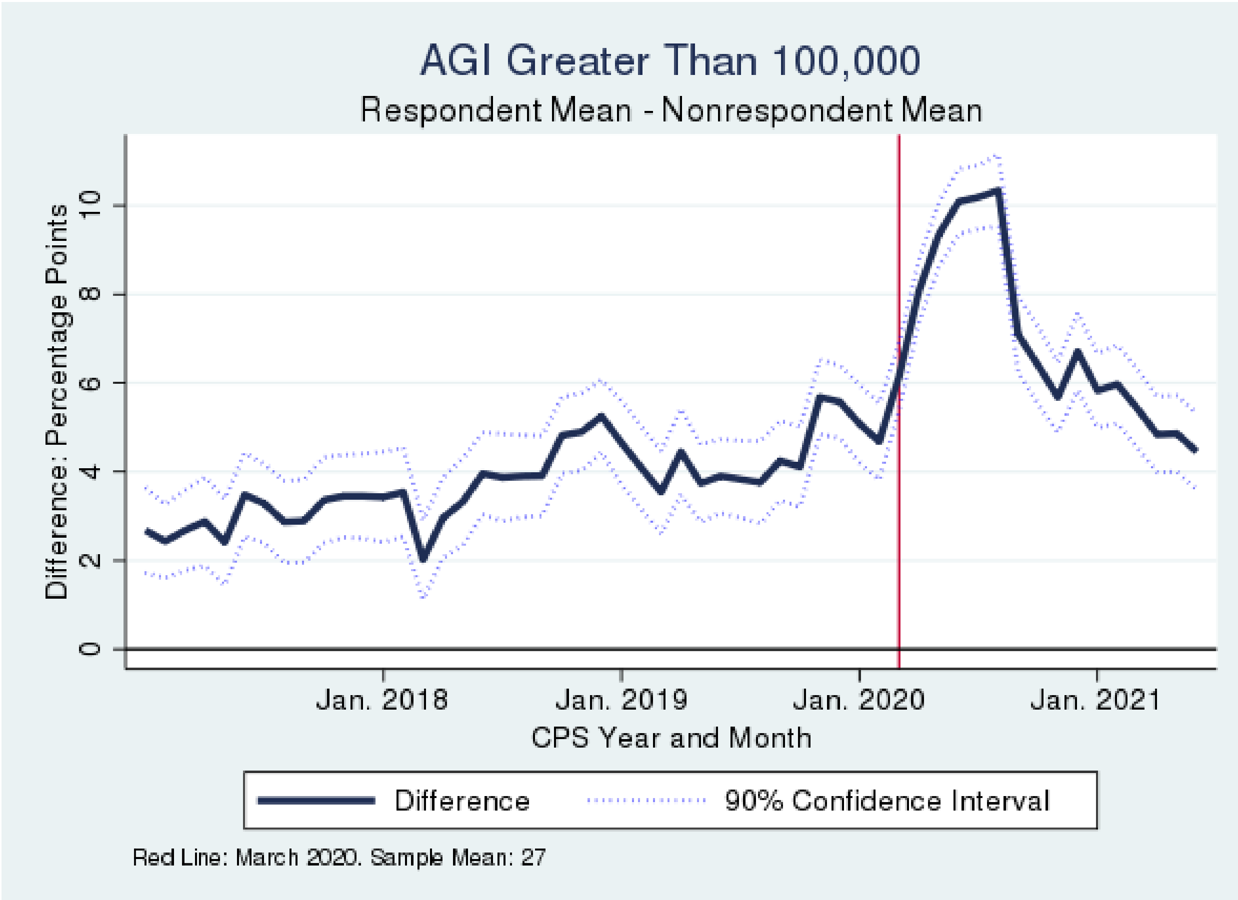
Source: Basic Monthly CPS matched to administrative data. Figure shows for each CPS month, the average for respondents minus the average for nonrespondents. CPS months used are January 2017 to June 2021.

Figure 6: Nonresponse Comparison for Adjusted Gross Income Less Than $25,000



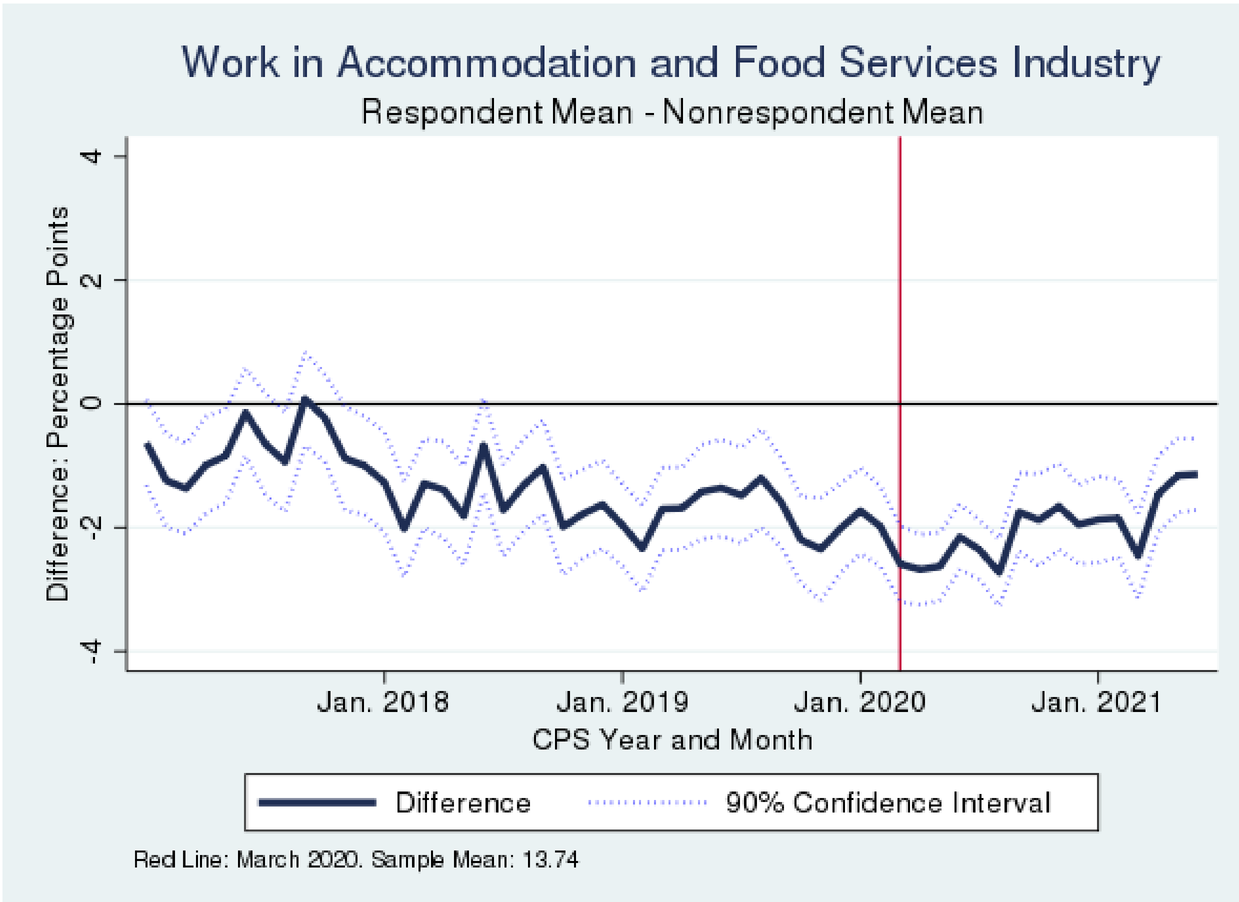
Source: Basic Monthly CPS matched to administrative data. Figure shows for each CPS month, the average for respondents minus the average for nonrespondents. CPS months used are January 2017 to June 2021.

Figure 7: Nonresponse Comparison for Adjusted Gross Income (AGI) Over $100,000



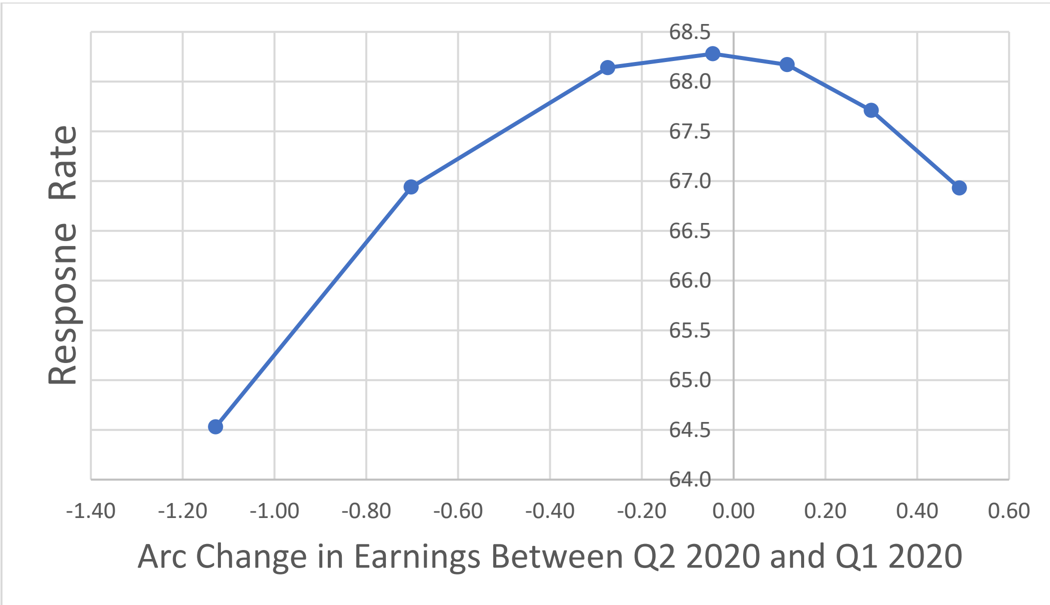
Source: Basic Monthly CPS matched to administrative data. Figure shows for each CPS month, the average for respondents minus the average for nonrespondents. CPS months used are January 2017 to June 2021.

Figure 8: Nonresponse Comparison for A Household Member Working in the Accommodation and Food Services Industry



Source: Basic Monthly CPS matched to administrative data. Figure shows for each CPS month, the average for respondents minus the average for nonrespondents. CPS months used are January 2017 to June 2021.

Figure 9: Response Rate and Earnings Changes, April-June 2020



Source: April, May, and June 2020 Basic Monthly CPS matched to administrative data. Figure shows how a household's change in administrative earnings from the LEHD between the 1st and 2nd Quarter of 2020 is related to a household's response rates. Sample consists only of households with positive earnings in the 1st and 2nd quarter. Estimates are from Kernel-weighted local polynomial smoothing with a Gaussian Kernel. Grid points are the 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles.

Figures 4 shows that respondent households are much more likely to have a household member over age 60. The difference is between 10 and 14 percentage points before the COVID-19 Pandemic and increases to about 16 points after March 2020. Figure 5 shows that in many months in the late 2010s, CPS respondents are no more likely to have a child under 10 than nonrespondents, but that respondents are less likely be in this category starting in 2020.

In terms of income, Figure 6 shows that respondents are no more likely to have Adjusted Gross Income (AGI) that is positive but less than $25,000 before March 2020, but after March 2020, respondents are about 4 percentage points less likely to be so, returning somewhat to pre-COVID levels in September 2020 when in-person interviewing resumed in many areas. Similarly, looking at whether the household's AGI is above $100,000 in Figure 7, respondents are about 2-6 percent points more likely to fall in this category before March 2020 with a slight time trend, but the difference jumps up after March 2020, with some fluctuation month-to-month. As seen in Figure 8, respondents are slightly less likely work in the accommodations and food service industry, with no notable change after March 2020.

To further explore the magnitude of nonresponse bias during the COVID-19 Pandemic, Figure 9 shows the relationship between response rates in the April-June 2020 CPS and change in earnings between the 1^{st} and 2^{nd} quarter of 2020. Households with a larger decrease in earnings have a lower response rate. Households at the 5^{th} percentile in change of earnings have a response rate of 64.5 percent, compared to a response rate of about 68 percent for the median household. However, this difference in response rates of 3.5 percent points is fairly small, potentially suggesting that there might not be a large difference between respondents and nonrespondents in unemployment status.

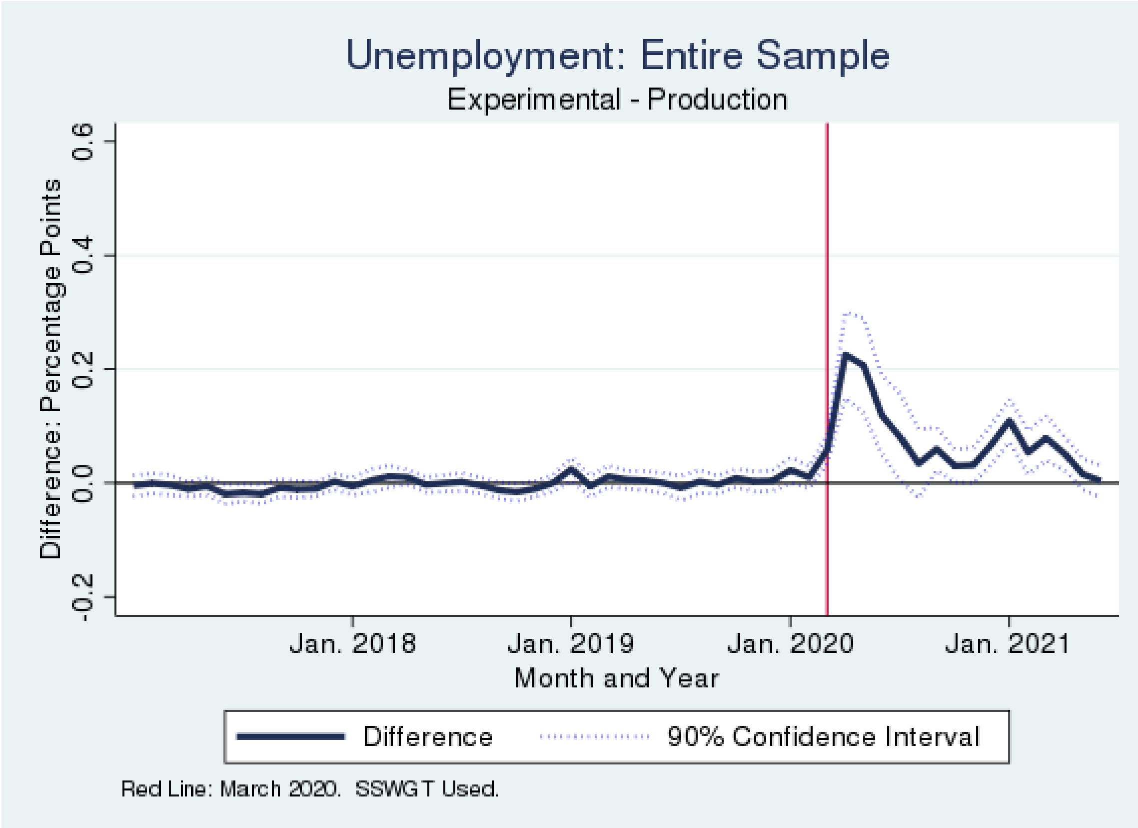
Overall, the results from Figures 4-9 show that there are differences between respondents and nonrespondents in the CPS. Respondents are much more likely to be older. And during the initial months of the pandemic, respondents appear to have higher income and less fluctuation in their earnings. How this translates into bias in the prior published labor force estimates depends on how well the geographic and demographic adjustments in the standard CPS weighting procedures reduce their differences. For example, because the weighted CPS age distribution is calibrated to age estimates from Census's Population Estimates Program, this likely reduces any nonresponse bias with respect to age. But if the results from Figure 4 are from differences in retirement status rather than age, then the age adjustments may not completely correct a bias in the labor force participation rate.

5.2. Change in Survey Estimates

Next, we show how estimates change when incorporating administrative data in the CPS weighting algorithm. We label the new administrative-based estimates as the "experimental" estimates and compare these to the estimates from the original weighting procedure without any seasonal adjustments. Figures 10 and 11 show the difference between the original and experimental estimates by CPS month for the national unemployment rate and labor force participation rate, respectively. Note that these estimates include all respondents age 15 and over, while the official employment statistics are based on individuals 16 and over. Additionally, due to computational limitations, the weights used in these exercises do not incorporate an additional process conducted by BLS intended to improve time series consistency.[9] Therefore, the employment statistics in this paper will not exactly match official estimates from the Bureau of Labor Statistics.

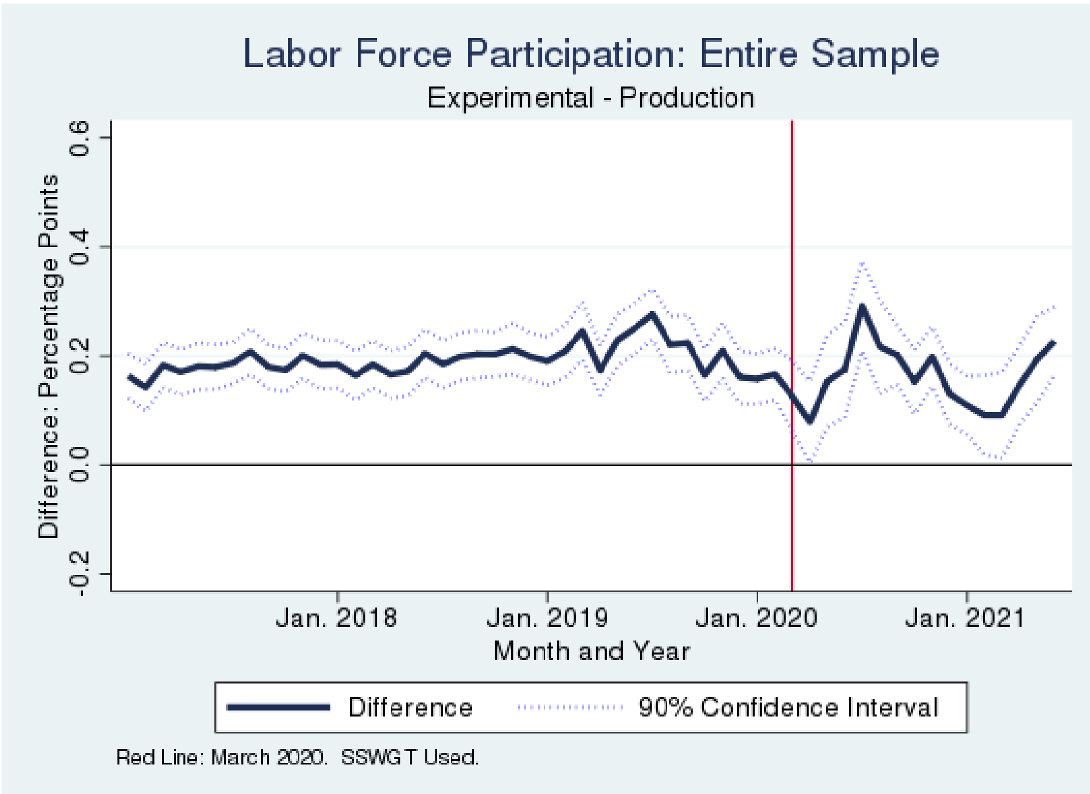
[9] That is, we use the second-stage weights rather than the composite weights. See McIllece (2020) for a discussion on how the COVID-19 pandemic affected estimates from the composite weights.

Figure 10 Change in Unemployment Rate Estimate



Source: Basic Monthly CPS matched to administrative data. Figure shows how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells. CPS months used are January 2017 to June 2021.

Figure 11 Change in Labor Force Participation Rate Estimate



Source: Basic Monthly CPS matched to administrative data. Figure shows how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells. CPS months used are January 2017 to June 2021.

Figure 10 shows that before March 2020, adding administrative data does not change the estimate of the unemployment rate. After March 2020, the experimental estimates of the unemployment rate are about 0.2 percentage points higher in April and May, with the difference decreasing after that. For the labor force participation rate in Figure 11, the experimental estimates are about 0.15 to 0.2 percentage points higher for most months prior to March 2020, and the magnitudes stay similar for some months after March 2020. While these differences are statistically significant, they are small, potentially suggesting minimal bias in the main published statistics. In particular, the unemployment rate typically has a margin of error of 0.2 percentage points, so this estimate of the bias in April and May of 2020 is in line with the magnitude of sampling error in a given month. Note that changing weighting algorithm can also affect the standard errors of the estimate (Little and Vartivarian 2005). Appendix Tables A.1 and A.2 show that the standard errors are similar across weighting procedures as well, evidence that adding the numerous administrative variables to the model did not negatively impact the precision of the estimates. Additionally, Appendix Figures A.1 and A.2 show that the estimates are also similar when using the administrative data that would have been available at the time the CPS data was being prepared.

Table 6 Change in Unemployment Rate Estimates by Demographics

	2017-2019		April-August 2020	
	Original Est.	Change Est.	Original Est.	Change Est.
Male	3.68	-0.006***	10.21	0.114***
Female	3.63	0.001***	11.91	0.160***
White	3.56	0.000***	10.63	0.108***
Black	6.79	-0.017***	15.19	0.256***
Hispanic	4.72	0.008***	14.79	0.160***
No High School	5.86	0.011***	16.29	0.137***
High School	4.15	-0.006***	12.81	0.103***
Some College	3.40	-0.007***	11.35	0.168***
College	2.20	-0.002***	6.98	0.078***

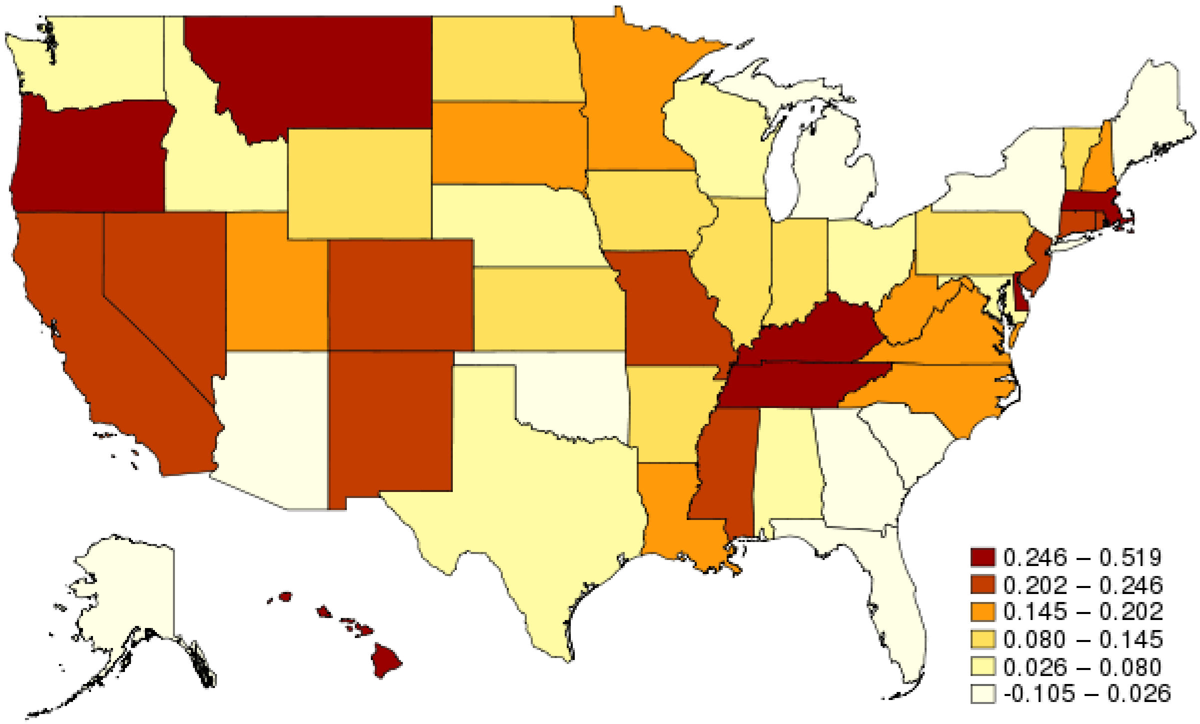
Source: Basic Monthly CPS matched to administrative data. Table shows by demographics how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells. Estimate for race and Hispanic origin are of individuals age 15 and over. Estimates for sex are of individuals age 20 and over. Estimates of education are of age 25 and over.

Table 7 Change in Labor Force Participation Rate Estimates by Demographics

	2017-2019		April-August 2020	
	Original Est.	Change Est.	Original Est.	Change Est.
Male	71.78	0.203***	69.62	0.240***
Female	58.87	0.210***	57.22	0.170***
White	62.34	0.200***	60.73	0.199***
Black	61.73	0.225***	59.20	0.172***
Hispanic	65.40	0.134***	63.40	0.137***
No High School	46.26	0.156***	44.01	0.192***
High School	57.94	0.265***	55.16	0.204***
Some College	65.69	0.297***	63.48	0.410***
College	73.98	0.202***	72.02	0.290***

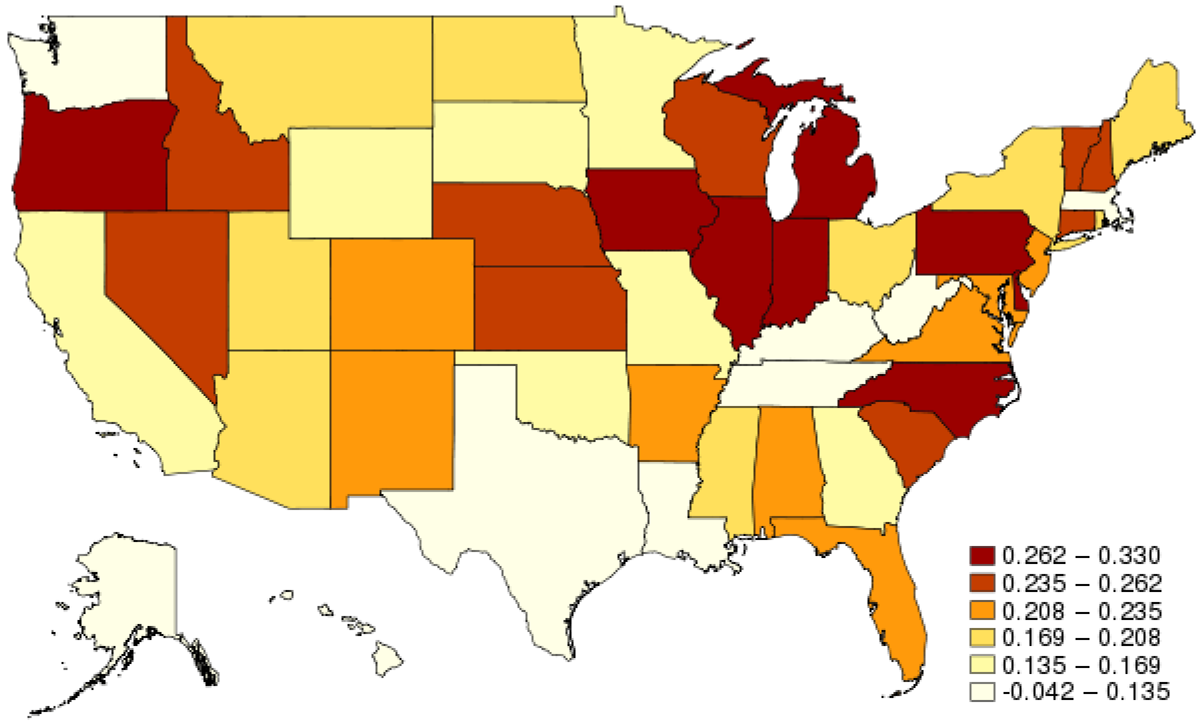
Source: Basic Monthly CPS matched to administrative data. Table shows by demographics how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells. Estimate for race and Hispanic origin are of individuals age 15 and over. Estimates for sex are of individuals age 20 and over. Estimates of education are of age 25 and over.

Figure 12: Change in Unemployment Rate Estimates by State: April to August 2020



Source: Basic Monthly CPS matched to administrative data. Figure shows by state how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells.

Figure 13: Change in Labor Force Participation Rate Estimates by State: January 2017 to February 2020



Source: Basic Monthly CPS matched to administrative data. Figure shows by state how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells.

To show how these effects vary by demographics, Tables 6 and 7 present differences in the unemployment and labor force participation rates by sex, race, ethnicity, and education. We show the averages from January 2017 to February 2020 as well as from April 2020 to August 2020 separately to highlight how COVID-19 may have impacted results. Overall, the findings are similar to the national results. For the unemployment rate, the increase in April-August 2020 is higher for Blacks at about 2.5 percentage points. For the labor force participation rate, the magnitude of the increase is U-shaped with respect to educational attainment, and respondents with some college or an associate degree have the largest increase in their estimate of the labor force participation rate. Finally, Figures 12 and 13 show the increase in the estimate of the unemployment rate from April 2020 to August 2020 by state, and Figure W2 shows the analogous changes in the labor force participation rate from January 2017 to

February 2020.[10] Overall, the results are similar to the national ones, with no discernable pattern across states.

6. Robustness Check with Entropy-Balance Weights

Although the additional administrative data should be the biggest driver of any change in estimates from the modified weighting process, modeling choices could also have an impact. In the CART-based weighting method, we modify only the initial household noninterview adjustment, not the second-stage calibration. Given we are not using administrative data in the second stage, there is potential that calibrating only to geographic and demographic estimates may counteract bias correction with respect to economic characteristics. Given this risk, we also run an entropy-balance weight (EBW) model with administrative data in both the household noninterview step and the second-stage step as a robustness check to verify the results are similar. EBW is similar to other minimization-based calibration approaches to weighting, such as the generalized regression estimator (GREG). The main difference is that EBW minimizes a log-distance function, while GREG estimators typically minimize a quadratic distance function.

The EBW procedure is described as follows, as described in Rothbaum et al. (2021). For a given CPS month, suppose we have n observations, where $i = 1, 2, \dots, n$ with base weights $q = \{q_1, q_2, \dots, q_n\}$, in our case determined by the sampling probability. Entropy balancing estimates the set of weights $w = \{w_1, w_2, \dots, w_n\}$ that solve the following minimization problem:

$$\min_{\mathrm{w}} \sum_{i=1}^{n} w_i \log\left(\frac{w_i}{q_i}\right) \qquad (1)$$

subject to several sets of constraints. First, we have J balance constraints, where $j = \{1, \dots, J\}$. Let $X = \{X_1, \dots, X_J\}$ be a matrix of observable characteristics. For each characteristic j, the balance constraint is defined to match a pre-specified constant $\bar{c}_j$, where:

$$\sum_{i=1}^{n} w_i c_j\left(X_{i,j}\right) = \bar{c}_j. \qquad (2)$$

[10] We graph the post-pandemic changes in the unemployment rate because there were more differences at the national level during this time period, using a different reference period for the labor force participation rate due to there being slightly more variation in the national-level difference in the post-pandemic period. Appendix Tables A.3-A.6 show the exact differences by state shown in these Figures, as well as the pre-pandemic state changes in the unemployment rate and post-pandemic changes in the labor force participation rate.

$c_j(\cdot)$ can be any arbitrary function.

Second, we have constraints on the weights themselves:

$$\sum_{i=1}^{n} w_i = \bar{w} \qquad (3)$$

$$w_i \geq 0. \qquad (4)$$

These ensure that the weights are non-negative and sum to some pre-specified total weight $\bar{w}$, which can be the population count or 1. The value of $\bar{w}$ does not affect the relative weights of each observation.

As such, the weights can be adjusted to match specified moments such as population means, variances, higher-order moments, moments of any transformed distribution of $X_{i,j}$, etc. In summary, entropy balancing adjusts the weights according to (1), subject to the constraints in (2), (3), and (4).

6.1. Household Noninterview Adjustment

We apply the EBW algorithm with a two-step procedure. First, we run the model at the household level with only administrative and geographic data. Essentially, this is a household noninterview adjustment using EBW rather than CART as described in Section 4. Let $X_{i,j}^{L}$ be characteristic j for household i, which can be a household-level measure from administrative data (e.g., whether anyone in the household has a W-2) or geography (e.g., whether the household is in a metropolitan area). For expositional purposes, assume that the weights are normalized to sum to 1, although in practice the weights will sum to population totals. We estimate the first-stage weights w_i^1 using entropy balancing, where the target on the right-hand side of constraint equation (2), the constant $\bar{c}_j$, is estimated from the sample of occupied housing units with base weights q_i:

$$\sum_{i \in R} w_i^1 c_j\left(X_{i,j}^{L}\right) = \sum_{i \in O} q_i c_j\left(X_{i,j}^{L}\right), \qquad (5)$$

where R is the set of all respondents and O is the set of all occupied housing units, which includes both respondents and nonrespondents. In other words, we use the set of occupied housing units linked to administrative data to create our targets, and then calibrate the weights of respondents to these targets.

6.2. Second Stage

After running the EBW process in the first stage, we run it again with additional constraints, this time at the person level. In this second stage, we create adjusted weights (denoted $w_{p,i}^2$ given individual p in household i and where $p = \{1, \dots, P\}$) to match externally estimated population controls while maintaining the moment conditions targeted in the first stage. In the first set of constraints related to the administrative data, we calculate per-person weighted moments from the first-stage weights. Given the number of people in household i, n_i^{HH}, we define the moment condition, with the balance constraints defined using the first-stage weights:

$$\sum_{p=1}^{P} w_{p,i}^2 \frac{1}{n_i^{HH}} c_j\left(X_{i,j}^L\right) = \sum_{i \in R} w_i^1 c_j\left(X_{i,j}^L\right). \tag{6}$$

As above, the term on the right-hand side is treated as the constant $\bar{c}_j$. This ensures that if we take the average weight of household members in household i (HH_i) as $\bar{w}_i^2 = \frac{1}{n_i^{HH}} \sum_{p \in HH_i} w_{p,i}^2$ the second-stage weights will satisfy the following condition:

$$\sum_{i \in R} \bar{w}_i^2 c_j\left(X_{i,j}^L\right) = \sum_{i \in R} w_i^1 c_j\left(X_{i,j}^L\right). \tag{7}$$

This does not require that $\bar{w}_i^2$ is equal to w_i^1 for any household i, just that the specified constraints from stage one hold in the final Entropy balance weights (EBW) weights, when the final weights are averaged across all household members. This procedure of dividing the household moments equally among the family members helps ensure that each person contributes to satisfying the moments from linked administrative data, which should reduce the variability of weights among household members. This can be particularly important for person-level statistics, such as poverty.

Next, we include an additional set of constraints to adjust the weights to match external estimates of race, Hispanic origin, age, and sex from Census's Population Estimate Program. This is analogous to the raking adjustment done in the production weighting algorithm. For a given demographic characteristic $X_{p,j}^D$ for individual p and external estimate $\bar{c}_j^D$, these moment conditions take the form:

$$\sum_{p=1}^{P^{RY}} w_{p,i}^2 c_j\left(X_{p,j}^D\right) = \bar{c}_j^D. \tag{8}$$

We calibrate to both the population estimates and the administrative data constraints at the same time. $w_{p,i}^2$ are then the final weights from this administrative data-based EBW algorithm.

6.3 Moments

With the EBW moments, we can include geographic, administrative, and population estimate program moments simultaneously. This offers a variety of benefits. For example, the EBW allows us to interact state indicators with our income variables, matching employment rate and UI administration differences at the state level. These interactions give valuable flexibility in addition to the improvements from the second-state calibration. We interact some of the administrative data moments with month-in-sample, further helping with balance given differential response rates by month-in-sample during 2020.

We calculate these moments separately for each replicate weight. This allows us to incorporate sampling uncertainty in the administrative data-based annual moments, capturing this form of error in our confidence intervals. For the LEHD variables, we include all four quarters of a year as moments to simplify our algorithm, allowing us to still incorporate this sub-annual information into our model.

6.3. Results

Table 8 Comparison of Administrative Data-Based Weighting for the CPS

May 2019

	Original		Adrec-CART		Adrec-EBW		P-Value Comparisons		
Statistics	Estimate	S.E.	Estimate	S.E.	Estimate	S.E.	Orig. vs CART	Orig. vs EBW	CART vs EBW
Unemployment	3.423	0.0911	3.427	0.0915	3.438	0.09306	0.661	0.414	0.561
L.F. Participation	62.13	0.1665	62.36	0.1634	62.3	0.1648	<.001	0.002	0.2
College Graduate	35.71	0.251	35.5	0.2509	35.66	0.2621	<.001	0.56	0.048
Married	49.87	0.2285	49.64	0.2187	49.4	0.2361	<.001	<.001	0.002
Citizen Native Born	83.41	0.1707	83.46	0.1719	83.77	0.1735	0.014	<.001	<.001
Non-Citizen	8.13	0.1377	8.15	0.1388	7.964	0.1391	0.186	<.001	<.001

May 2020

	Original		Adrec-CART		Adrec-EBW		P-Value Comparisons		
Statistics	Estimate	S.E.	Estimate	S.E.	Estimate	S.E.	Orig. vs CART	Orig. vs EBW	CART vs EBW
Unemployment	13.04	0.2001	13.24	0.1967	13.35	0.2068	<.001	<.001	0.094
L.F. Participation	60.02	0.179	60.17	0.1809	59.92	0.1801	0.003	0.205	<.001
College Graduate	37.77	0.2899	37.3	0.2888	36.65	0.3021	<.001	<.001	<.001
Married	51.11	0.2293	50.62	0.2356	49.32	0.2754	<.001	<.001	<.001
Citizen Native Born	83.85	0.1726	83.68	0.1724	84.09	0.1717	<.001	0.007	<.001
Non-Citizen	7.443	0.1558	7.644	0.1612	7.495	0.1628	<.001	0.401	0.015

Table compares estimates from Basic Monthly CPS using various weighting methodology. “Original” is the second stage weights from the existing production method. “Adrec-CART” is the administrative data method as described in Section 4 which used CART to create new cells for the household noninterview adjustment but leaves the rest of weighting algorithm the same. “Adrec-EBW” is the administrative data method described in Section 6 which use entropy-balancing to create weights, replacings all steps of the existing CPS weighting algorithm, including the second-stage steps which calibrate to estimates from Census’s Population Estimate Program. The estimate for the percentage of college graduates includes only individuals over 25. All other estimates are conditional on the respondent being over 15.
Source: Basic Monthly CPS matched to administrative data

Figure 14: Change in Unemployment Rate Estimates from Administrative-Based Weighting Methods

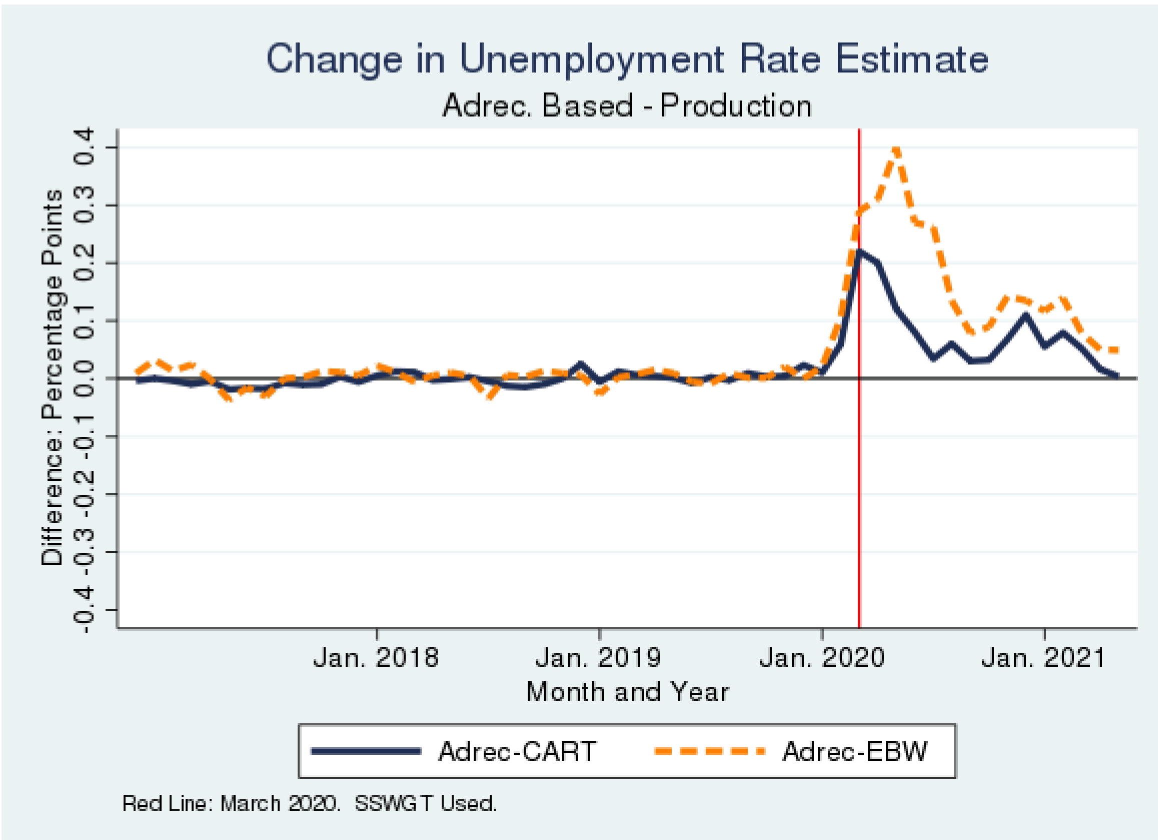


Figure presents difference between the production estimate and the administrative data-based estimate (bottom panel). See Table 8 for more details.
Source: Basic Monthly CPS matched to administrative data

Figure 15: Change in Labor Force Participation Rate Estimates from Administrative-Based Weighting Methods

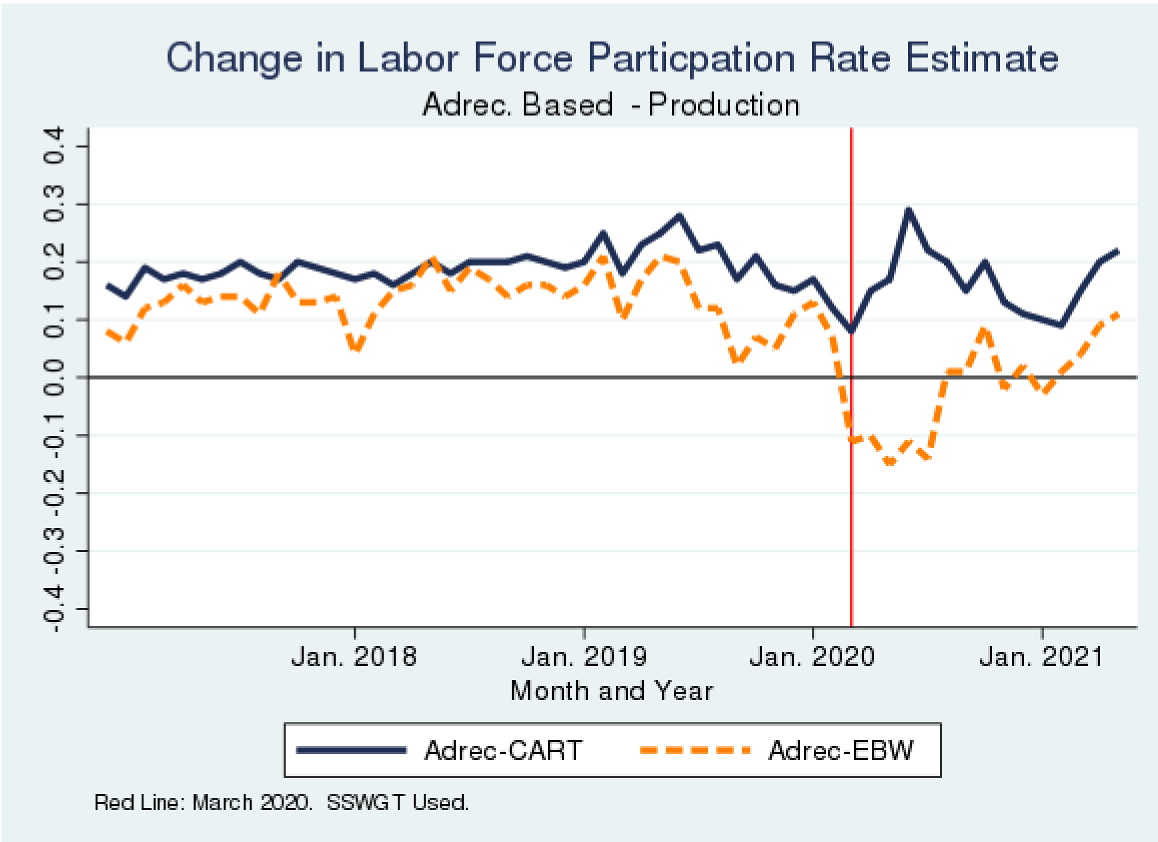


Figure presents difference between the production estimate and the administrative data-based estimate (bottom panel). See Table 8 for more details.

Source: Basic Monthly CPS matched to administrative data

Table 8 and Figures 14 and 15 present statistics comparing estimates from the EBW algorithm to the both the original estimates and the CART-based method. Some of these differences are statistically significant despite small differences in point values because the estimates are all run on the exact same set of observations, making the estimates highly correlated. In addition to showing statistics for unemployment and labor force participation, we show how the estimates of college graduation, marriage, citizen native-born, and non-citizen proportions change with the weighting methods in Table 8, with additional estimates by each month from January 2017 to June 2021 in Appendix Table A.7 and Appendix Figures A.3-A.6. While the CART method was designed to minimize bias with respect to labor force status and not the other measures, we still look at some non-employment statistics in order to get a sense on the sensitivity to alternative weighting approaches. For statistical testing, we look only at the May 2019 and May 2020 months (picking one month pre- and post-COVID given computational constraints for the demanding EBW model).

Overall, the results with this alternative weighting methodology are very similar to those from our main strategy. Under both weighting strategies, estimates are essentially identical before COVID, with some minor differences emerging thereafter, and gaps between the two new weighting strategies are in line with the typical margin of error of unemployment rate estimates. This suggests two key conclusions. First, the addition of administrative data changes CPS employment estimates in a consistent manner. Second, if we interpret the differences from the official weights as an approximation of nonresponse bias in CPS estimates, then augmenting the weighting process with the new administrative records is robust to the choice of weighting algorithm.

7. Comparisons to Prior Papers

If we interpret the difference between the weighting measures as capturing nonresponse bias in the unemployment rate, our measure of the bias is much lower than prior estimates from Heffetz and Reeves (2019,2021) and Krueger, Mas, and Niu (2017). In this section, we discuss how our results can be reconciled.

7.1 Late Response Analysis

First, we consider the approximation of the bias in the CPS using a continuum of resistance model. Heffetz and Reeves (2019,2021) estimate the bias by comparing the unemployment rate of CPS respondents who respond within one or two contact attempts versus after three or more contact attempts. This model proposes that nonrespondents and respondents who require more contact attempts to interview are similar. Heffetz and Reeves (2019,2021) find that respondents who take multiple contact attempts have a lower unemployment rate and a higher labor force participation rate, so nonrespondents may have a lower unemployment rate as well. Prior research has cast doubt on whether this is a valid way to approximate the characteristics on nonrespondents (e.g., Lin and Schaeffer 1995), but given our access to administrative data on CPS nonrespondents, we can directly examine whether the continuum of resistance model is appropriate in the CPS.

Table 9 Validity of Continuum of Resistance Model in the CPS data

	Respondents: Number of Contact Attempts			Nonrespondents
	1	2	3 or More	
Percent Household with a Member Who Has a W2 or 1040-SE	65.6	70.4	74.4	70.59
Percent Household with a Member Who Does Not Have a W2 or 1040-SE	48.4	42.9	36.14	33.08
Percent Household with a Member Who Has OASDI or SSI SSA Benefits	42.3	36.3	27.9	23.84

Source: 2017-2021 Basic Monthly CPS matched to administrative data.

Table 9 shows how our administrative data measures described in Section 3 vary by number of contact attempts among respondents and nonrespondents. Households that require more contact attempts are more likely to be employed, and nonrespondents are overall more like households that require more contact attempts than ones needing only one attempt. For example, 42.3% of households with one contact attempt have a household member who receives social security benefits, but this drops to 27.9% for households that require three or more attempts and drops further to 23.84% for nonrespondents. The differences are not as large when looking at households with any W2 or 1040 schedule SE, where the percentage of nonrespondents with these tax forms (70.59%) is not significantly different from the percentage for households with two attempts (70.4%). But even here, nonrespondent households have higher rates on employment than households with one contact attempt.

We conclude that there is evidence of a continuum of resistance model in the CPS, at least for measures related to labor force participation. Given that we find a higher estimate of the labor force participation rate with administrative data, the direction of our estimate of the bias is the same as in Heffetz and Reeves (2019). However, the magnitude of the bias we estimate is small. There are several possible explanations. While Heffetz and Reeves (2019) find differences between late and early respondents after controlling for demographic characteristics, the CPS weighting procedures also make use of geographic adjustments that explain within-sample differences in labor force characteristics. In addition, the difference between late and early responders may be mitigated by CPS' relatively high response rate of around 85% in the pre-pandemic period. More formally, a fundamental decomposition of nonresponse bias is

$$E(Y \mid R) - E(Y) = (1 - P(R))(E(Y|R) - E(Y|R = 0))$$

where Y is the outcome of interest and R is the indicator of response. It follows that even if the difference between respondents and nonrespondents is moderately high, if the nonresponse rate is low, then the resulting nonresponse bias will be low as well. In summary, the sign and magnitude of the bias we find for the labor force participation rate in the pre-pandemic period appears to be consistent with Heffetz and Reeves (2019).

In contrast to labor force participation, our unemployment results differ from Heffetz and Reeves' (2019). In the pre-pandemic period, we do not find any measurable bias. Given Heffetz and Reeves (2019) find smaller differences for unemployment compared to labor force participation (1.5 percentage points for unemployment compared to 4.9 percent for labor force participation), this could mean any actual bias is too small to detect given CPS' sample size. And for months in the spring of 2020, we find that the administrative data-based weights result in a higher estimate of the unemployment rate, where the continuum of resistance method suggest that the bias should go in the opposite direction. However, the nature of the survey response process was very different due to pandemic-related restrictions. And as discussed in Section 2.2, noncontact bias from phone number quality in the Census contact frame may have led to more high-income people being successfully contacted, counteracting any pre-existing bias in contactability related to employment.

7.2 Rotation Group Bias

Our estimates of the nonresponse bias in the unemployment rate are in the opposite direction of what is suggested by rotation group bias. Bailar (1975) was the first to document that the unemployment rate is higher for the first and fifth rotation groups than for other groups. This is surprising, given each rotation group is designed to be nationally representative by itself. Krueger, Mas, and Niu (2017) examine rotation group bias through the 2010s and find that it increased after the 1994 CPS redesign, which introduced Computer-Assisted Personal Interviews (CAPI). They document that the average unemployment rate from 2009 to 2013 was 9.3 percent for respondents in the first rotation group compared to 8.3 percent for respondents in the eighth rotation group. Heffetz and Reeves (2021) use this rotation group analysis to estimate that the April 2020 unemployment rate may have been underestimated by 1.6 percentage points, much larger than our estimates.

While rotation group bias is evident from the CPS data, its causes are unclear. It could be nonresponse bias in which the characteristics of the respondent sample change between rotation groups. Or it could be measurement error, where either being interviewed in the CPS in the past month affects how people respond to the survey or the higher prevalence of phone interviews in later months introduces some mode effect. Krueger, Mas, and Niu (2017) provide evidence that rotation group bias is caused by nonresponse, noting that it has been increasing over time along with nonresponse rates, that rotation group bias is higher for demographic groups that also have higher nonresponse rates, and that there is little rotation group bias for households interviewed for all eight months. They also rule out other measurement error explanations, such as long-term unemployed people changing their response behavior. However, given that Krueger, Mas, and Niu (2017) examine nonresponse bias through indirect means, there could be alternative explanations. For example, Kreuter, Müller, and Trappmann (2010) find that late responders to surveys had higher measurement error in their responses. Therefore, if a certain group in the CPS has lower response rates, there may be also some type of measurement error process that is causing them to exhibit larger rotation group bias.

Table 10: Change in Rotation Group Bias with Experimental Weights: 2017-2019

	Rotation Group									
Weights	1	2	3	4	5	6	7	8	Mean	Slope
Original	112.2	105	98.96	97.3	99.96	96.06	95.99	95.03	4.03	-2.008
Experimental	112.2	104	98.9	97.28	100.1	96.03	96.08	95.04	4.025	-1.992

Source: Basic Monthly CPS matched to administrative data.

Table 11: Change in Rotation Group Bias with Experimental Weights: April to August 2020

	Rotation Group									
Weights	1	2	3	4	5	6	7	8	Mean	Slope
Original	103.1	104	103.6	104.6	96.44	94.07	96.11	98.25	11.55	-1.3
Experimental	102.5	104	103.3	105.1	97.16	94	95.99	98.05	11.68	-1.268

Source: Basic Monthly CPS matched to administrative data.

To further explore whether rotation group bias is due to measurement error or nonresponse bias, Tables 10 and 11 show how rotation group bias changes between the original and our experimental weights. We pool 2017 to 2019 and April 2020 to August 2020 separately. We adopt the same methodology as Krueger, Mas, and Niu (2017) wherein we estimate the unemployment rate for each rotation group, divide it by the pool average, and multiply by 100. The key estimate of rotation group bias is the slope from a linear regression of the unemployment rate on the rotation group number. We find no significant change in the magnitude of rotation group bias with the new experimental weights. These estimates suggest that if our experimental weights are properly accounting for differential nonresponse bias across rotation groups, then rotation group bias is primarily due to measurement error, not nonresponse bias. Further research is needed to determine what measurement error process is causing rotation group bias.

7.3 Results from Other Countries

The aforementioned papers look at the United States, but many European countries have population registers and more centralized administrative data. Some countries, such as the Netherlands (Houbiers 2004) and Denmark (Statistics Denmark n.a.) use these registers for sampling and weighting in their labor force surveys. Notably, Austria in 2014 transitioned to using administrative data in weighting for their labor force survey, the Austrian microcensus. Meraner et al. (2016) find that changes in the weighting procedures resulted in a decrease in the estimates of the employment rate and labor force participation rate and an increase in the estimate of the unemployment rate. This implies that employed people were more likely to respond to the survey, which is like what we find in the CPS post-COVID. Besides cultural differences, there are other survey differences they may explain the results, such as the Austrian survey being mandatory but the U.S. CPS being voluntary. Nevertheless, the opposite direction of bias in our analyses compared to Meraner et al. (2016) is evidence that the magnitude and direction of nonresponse bias in employment statistics may vary across settings.

8. Conclusion

In this paper, we present findings on how adding administrative data to the CPS weighting algorithm changes employment estimates from the survey. With our primary CART-based algorithm, any differences in the unemployment and labor force participation rates are at most 0.2 percentage points, comparable to the typical margin of error in the unemployment rate estimate. Some of the unemployment differences are higher with the entropy-balance weighting algorithm, but any increase in the unemployment rate is at most 0.4 percentage points. And overall, the results are similar across demographics except for some differences by educational attainment.

In the weighting algorithm, we add numerous high-quality administrative data sources from the IRS, state employment agencies, SSA, and other sources. This gives us valuable information on nonrespondents, which should mean that the administrative data-based weights are better able to correct for any nonresponse bias in the CPS. However, note that the concept of employment status inferred from the administrative data are not the same as the concepts measured in the CPS, so the administrative data may not be able to correct for all nonresponse with respect to employment characteristics Our administrative data through SSA program benefit data and retirement income receipt provides accurate indicators of someone being out the labor force for an extended period of time, such as due to disability or retirement. On the other hand, our quarterly earnings data provide good information on someone with a loss of employment at the start of the COVID-19 pandemic but does not provide data on whether they were looking for work during this time or the precise weeks they were working.

Overall, the fact that we find minimal changes in employment statistics even after adding numerous administrative datasets to the weighting model is suggestive that nonresponse bias in employment statistics is minimal in the CPS. This is encouraging for one aspect of data quality, although other sources of error could create bias; the continued existence of rotation group bias and the large increase in the number of people reporting absent from work for “other reasons” during the COVID-19 pandemic (Bureau of Labor Statistics 2020) are examples of such concerns. Our reweighting exercise suggests that these facets of the data are not due to a lack of a representative sample, but potentially other types of errors, such as measurement error. For future work, continuing to evaluate how CPS estimates change with modified weighting procedures will be vital to detect possible nonresponse bias that could start to manifest. For example, if an online response option were implemented for the CPS, that could have the potential to change the characteristics of the respondent sample.

References

The American Association for Public Opinion Research. 2016. "Standard Definitions: Final Dispositions of Case Codes and Outcome Rates for Surveys. 9th edition." AAPOR. Available at https://www.aapor.org/Standards-Ethics/Standard-Definitions-(1).aspx.

Bailar, B. 1975. “The Effects of Rotation Group Bias on Estimates from Panel Surveys”. Journal of the American Statistical Association, 70, 23±30. Available at https://www.jstor.org/stable/2285370.

Bee, C. Adam, Graton MR Gathright, and Bruce D. Meyer. 2015. "Bias from unit non-response in the measurement of income in household surveys." Unpublished paper, University of Chicago.

Czajka, John L., and Amy Beyler. 2016. *Background Paper Declining Response Rates in Federal Surveys: Trends and Implications*. Washington: Mathematica Policy Research. Available at https://aspe.hhs.gov/system/files/pdf/255531/Decliningresponserates.pdf.

Eggleston, Jonathan. 2021. "Survey Isolation During COVID-19: The Effects of Suddenly Relying on Address-Matched Phone Numbers for Interviewing Households." *American Association of Public Opinion Researchers Annual Conference.*

Eggleston, Jonathan, and Ashley Westra. 2020. "Incorporating Administrative Data in Survey Weights for the Survey of Income and Program Participation." *SIPP Working Paper* No. 292. U.S. Census Bureau. https://www.census.gov/content/dam/Census/library/working-papers/2020/demo/SEHSD-WP2020-07.pdf.

Groves, Robert M., and Emilia Peytcheva. 2008. "The Impact of Nonresponse Rates on Nonresponse Bias: A Meta-Analysis." *Public Opinion Quarterly* 72, no. 2: 167-189. Available at https://doi.org/10.1093/poq/nfn011.

Heffetz, Ori, and Daniel B. Reeves. 2019. "Difficulty of reaching respondents and nonresponse Bias: Evidence from large government surveys." *Review of Economics and Statistics* 101, no. 1: 176-191. https://doi.org/10.1162/rest_a_00748.

Heffetz, Ori, and Daniel Reeves. 2021. Measuring Unemployment in Crisis: Effects of COVID-19 on Potential Biases in the CPS. No. w28310. National Bureau of Economic Research. https://www.nber.org/system/files/working_papers/w28310/w28310.pdf.

Houbiers, Marianne. "Towards a social statistical database and unified estimates at Statistics Netherlands." Journal of official statistics 20, no. 1 (2004): 55. https://www.scb.se/contentassets/f6bcee6f397c4fd68db6452fc9643e68/towards-a-social-statistical-database-and-unified-estimates-at-statistics-netherlands.pdf.

Kreuter, Frauke, Gerrit Müller, and Mark Trappmann. "Nonresponse and measurement error in employment research: making use of administrative data." *Public Opinion Quarterly* 74, no. 5 (2010): 880-906. https://doi.org/10.1093/poq/nfq060.

Krueger, Alan B., Alexandre Mas, and Xiaotong Niu. 2017. The Evolution of Rotation Group Bias: Will the Real Unemployment Rate Please Stand Up?" *Review of Economics and Statistics*, 2(99): 258-264. https://doi.org/10.1162/REST_a_00630.

Lin, I-Fen, and Nora Cate Schaeffer. "Using survey participants to estimate the impact of nonparticipation." Public Opinion Quarterly 59, no. 2 (1995): 236-258. https://doi.org/10.1086/269471.

Little, Roderick J., and Sonya Vartivarian. 2005. "Does weighting for nonresponse increase the variance of survey means?" *Survey Methodology* 31, no. 2: 161-168. Available at https://www150.statcan.gc.ca/n1/pub/12-001-x/2005002/article/9046-eng.pdf.

Meraner, Angelika, Daniela Gumprecht, and Alexander Kowarik. 2016. "Weighting procedure of the austrian microcensus using administrative data." *Austrian Journal of Statistics* 45, no. 3: 3-14.

McIllece, Justin. 2020. "COVID-19 and the Current Population Survey: Response Rates and Estimation Bias." *US Bureau of Labor Statistics* 20 (2020). https://www.bls.gov/osmr/research-papers/2020/st200030.htm

Rothbaum, Jonathan, and Adam Bee. 2020. "Coronavirus Infects Surveys, Too: Nonresponse Bias During the Pandemic in the CPS ASEC." *Social, Economic, and Housing Statistics Division Working Paper #2020-10*. Available at https://www.census.gov/library/working-papers/2020/demo/SEHSD-WP2020-10.html.

Rothbaum, Jonathan, Jonathan Eggleston, Adam Bee, Mark Klee, and Brian Mendez-Smith. 2021. "Addressing Nonresponse Bias in the American Community Survey During the Pandemic Using Administrative Data." 2021 AMERICAN COMMUNITY SURVEY RESEARCH AND EVALUATION REPORT MEMORANDUM SERIES # ACS21-RER-05 and SEHSD Working Paper #2021-24, U.S. Census Bureua. https://www.census.gov/library/working-papers/2021/acs/2021_Rothbaum_01.html

Statistics Netherlands. n.d. "Labor Force Survey." Accessed January 19, 2022. https://www.dst.dk/en/Statistik/dokumentation/metode/aku-arbejdskraftundersoegelsen.

U.S. Bureau of Labor Statistics. 2021. "Impact of the coronavirus (COVID-19) pandemic on The Employment Situation for October 2021." Accessed November 29, 2021. https://www.bls.gov/covid19/employment-situation-covid19-faq-october-2021.htm.

U.S. Census Bureau. 2019. "Current Population Survey Design and Methodology Technical Paper 77." October 2019. U.S. Census Bureau. https://www2.census.gov/programs-surveys/cps/methodology/CPS-Tech-Paper-77.pdf.

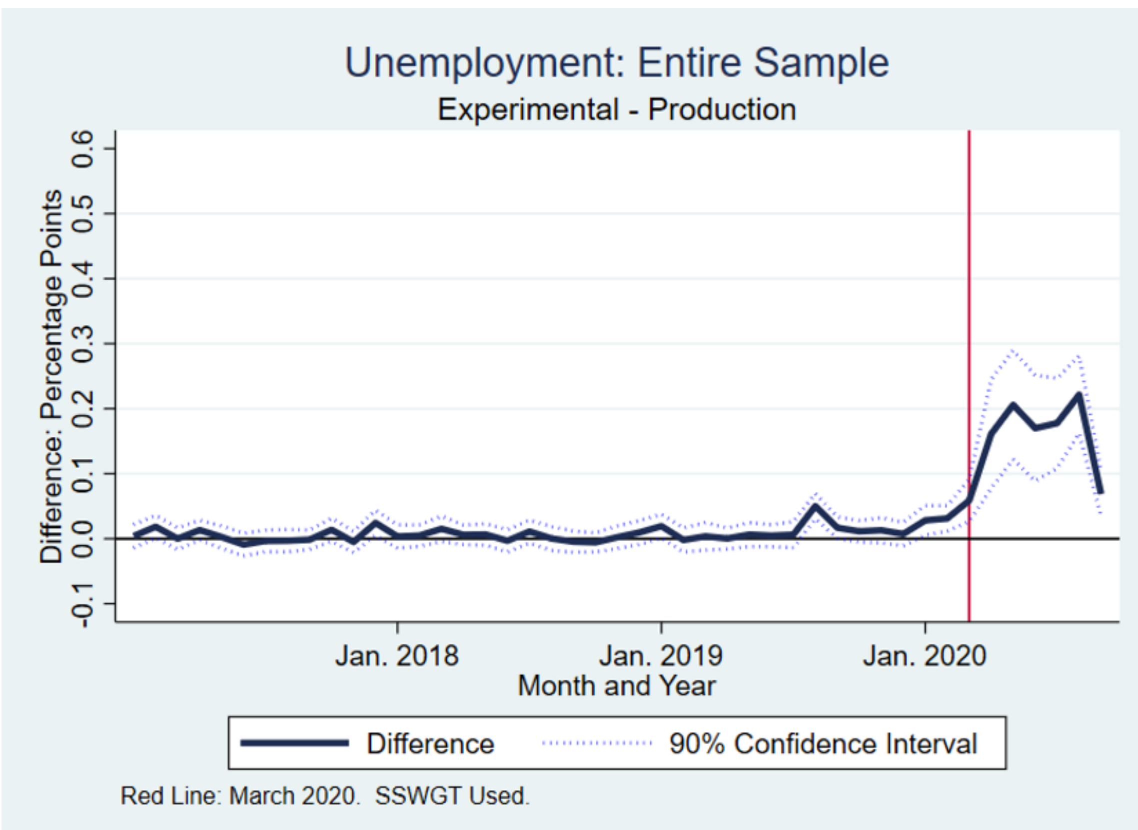
U.S. Census Bureau. 2020. "Census Bureau Resumes In-Person Interviews for Ongoing Surveys." U.S. Census Bureau press release, July 1, 2020. On the Census website. https://www.census.gov/newsroom/press-releases/2020/ongoing-surveys.html, accessed 28 July 2022.

Appendix

A.1 Lagged Administrative Data

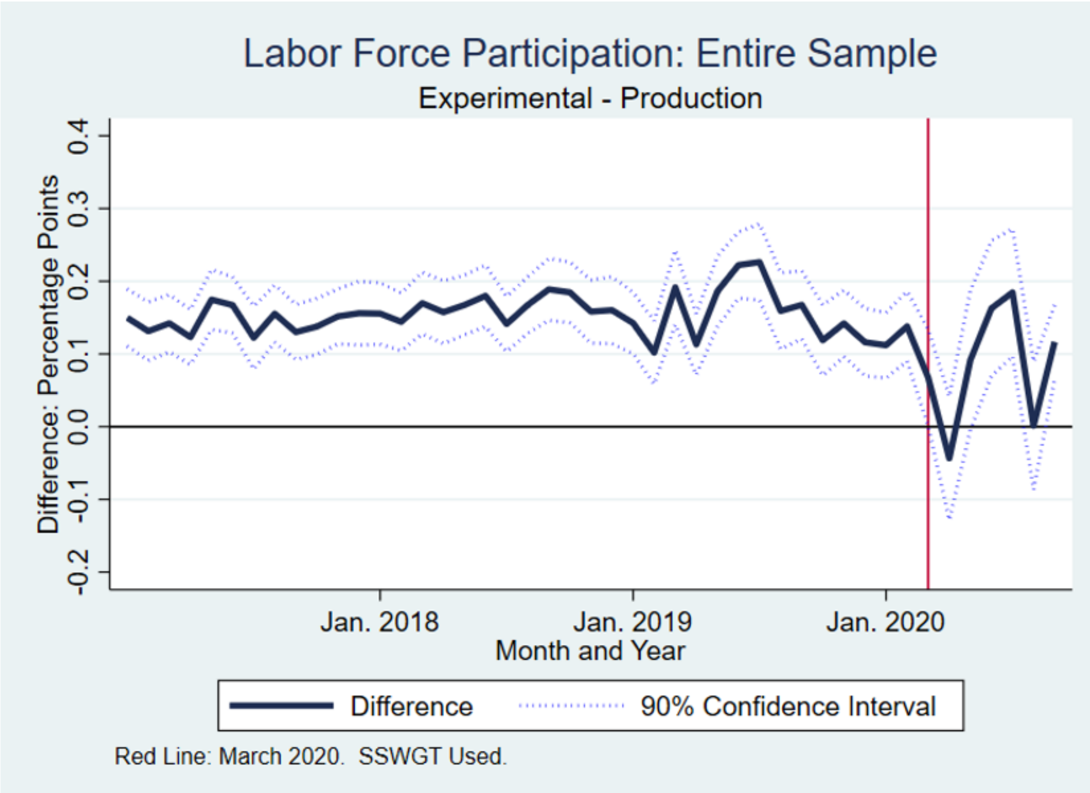
The results in the body of this paper use administrative data with the same reference year as the CPS. This means that the administrative used did not become available until well after the CPS was released—a year or more depending on the CPS month. We did this to use the administrative data that was most correlated with labor force status, providing the best possible approximations for nonresponse bias. In earlier iterations of this project, we used administrative data that would have been available at the time of CPS production. Below are two national-level figures showing the changes in estimates with these lagged administrative data. Overall, the estimates are very similar.

Figure A.1 Change in Unemployment Rate Estimate-Lagged Administrative Data



Source: Basic Monthly CPS matched to lagged administrative data (administrative data that would have been available at the time of CPS production). Figure shows how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells. CPS months used are January 2017 to June 2020.

Figure A.2 Change in Labor Force Participation Rate Estimate



Source: Basic Monthly CPS matched to lagged administrative data (administrative data that would have been available at the time of CPS production). Figure shows how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells. CPS months used are January 2017 to June 2020.

A.2 Additional Tables and Figures

Table A.1 Unemployment Rate Estimates: Original and CART-based Method

		Original		Adrec-CART		Difference	
Year	Month	Est.	S.E.	Est.	S.E.	Est.	S.E.
2017	1	0.0517	0.0012	0.0517	0.0012	0.0000	0.0001
2017	2	0.0498	0.0011	0.0498	0.0012	0.0000	0.0001
2017	3	0.0461	0.0010	0.0460	0.0010	0.0000	0.0001
2017	4	0.0417	0.0011	0.0416	0.0011	-0.0001	0.0001
2017	5	0.0415	0.0010	0.0415	0.0010	-0.0001	0.0001
2017	6	0.0457	0.0010	0.0455	0.0010	-0.0002	0.0001
2017	7	0.0468	0.0010	0.0466	0.0010	-0.0002	0.0001
2017	8	0.0462	0.0010	0.0460	0.0010	-0.0002	0.0001
2017	9	0.0410	0.0010	0.0409	0.0010	-0.0001	0.0001
2017	10	0.0395	0.0010	0.0394	0.0010	-0.0001	0.0001
2017	11	0.0395	0.0010	0.0394	0.0010	-0.0001	0.0001

2017	12	0.0396	0.0010	0.0397	0.0010	0.0000	0.0001
2018	1	0.0457	0.0010	0.0456	0.0010	-0.0001	0.0001
2018	2	0.0444	0.0011	0.0444	0.0012	0.0000	0.0001
2018	3	0.0417	0.0011	0.0418	0.0011	0.0001	0.0001
2018	4	0.0374	0.0010	0.0376	0.0010	0.0001	0.0001
2018	5	0.0361	0.0009	0.0361	0.0008	0.0000	0.0001
2018	6	0.0427	0.0010	0.0427	0.0010	0.0000	0.0001
2018	7	0.0414	0.0010	0.0414	0.0010	0.0000	0.0001
2018	8	0.0403	0.0010	0.0403	0.0010	0.0000	0.0001
2018	9	0.0362	0.0009	0.0361	0.0009	-0.0001	0.0001
2018	10	0.0362	0.0010	0.0360	0.0010	-0.0002	0.0001
2018	11	0.0353	0.0009	0.0352	0.0009	-0.0001	0.0001
2018	12	0.0374	0.0009	0.0374	0.0009	0.0000	0.0001
2019	1	0.0449	0.0011	0.0451	0.0011	0.0002	0.0001
2019	2	0.0416	0.0010	0.0415	0.0010	-0.0001	0.0001
2019	3	0.0397	0.0010	0.0398	0.0010	0.0001	0.0001
2019	4	0.0338	0.0009	0.0339	0.0009	0.0001	0.0001
2019	5	0.0342	0.0009	0.0343	0.0009	0.0000	0.0001
2019	6	0.0392	0.0011	0.0392	0.0011	0.0000	0.0001
2019	7	0.0407	0.0011	0.0406	0.0011	-0.0001	0.0001
2019	8	0.0387	0.0009	0.0387	0.0009	0.0000	0.0001
2019	9	0.0338	0.0009	0.0338	0.0009	0.0000	0.0001
2019	10	0.0337	0.0010	0.0338	0.0010	0.0001	0.0001
2019	11	0.0334	0.0008	0.0335	0.0008	0.0000	0.0001
2019	12	0.0341	0.0009	0.0342	0.0009	0.0000	0.0001
2020	1	0.0405	0.0010	0.0407	0.0009	0.0002	0.0001
2020	2	0.0387	0.0010	0.0388	0.0010	0.0001	0.0001
2020	3	0.0462	0.0010	0.0467	0.0011	0.0006	0.0002
2020	4	0.1456	0.0021	0.1478	0.0021	0.0023	0.0005
2020	5	0.1304	0.0020	0.1324	0.0020	0.0021	0.0005
2020	6	0.1124	0.0019	0.1136	0.0019	0.0012	0.0004
2020	7	0.1041	0.0017	0.1049	0.0017	0.0008	0.0005
2020	8	0.0847	0.0017	0.0850	0.0017	0.0003	0.0004
2020	9	0.0763	0.0015	0.0769	0.0015	0.0006	0.0002
2020	10	0.0657	0.0013	0.0660	0.0013	0.0003	0.0002
2020	11	0.0636	0.0013	0.0639	0.0013	0.0003	0.0002
2020	12	0.0650	0.0014	0.0657	0.0015	0.0007	0.0002
2021	1	0.0687	0.0014	0.0698	0.0014	0.0011	0.0002
2021	2	0.0667	0.0014	0.0673	0.0014	0.0005	0.0002
2021	3	0.0623	0.0015	0.0631	0.0014	0.0008	0.0002
2021	4	0.0582	0.0013	0.0587	0.0013	0.0005	0.0002
2021	5	0.0566	0.0014	0.0568	0.0013	0.0002	0.0002
2021	6	0.0622	0.0013	0.0623	0.0013	0.0000	0.0002

Source: Basic Monthly CPS matched to administrative data. Table show how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells. CPS months used are January 2017 to June 2021.

Table A.2 Labor Force Participation Rate Estimates: Original and CART-based Method

		Original		Adrec-CART		Difference	
Year	Month	Est.	S.E.	Est.	S.E.	Est.	S.E.
2017	1	0.6174	0.0015	0.6190	0.0015	0.0016	0.0002
2017	2	0.6207	0.0017	0.6221	0.0017	0.0014	0.0003
2017	3	0.6212	0.0017	0.6231	0.0017	0.0018	0.0003
2017	4	0.6222	0.0017	0.6239	0.0017	0.0017	0.0002
2017	5	0.6217	0.0016	0.6235	0.0016	0.0018	0.0003
2017	6	0.6263	0.0018	0.6280	0.0018	0.0018	0.0002
2017	7	0.6296	0.0018	0.6314	0.0017	0.0019	0.0002
2017	8	0.6236	0.0018	0.6256	0.0018	0.0021	0.0003
2017	9	0.6237	0.0017	0.6255	0.0017	0.0018	0.0002
2017	10	0.6198	0.0017	0.6215	0.0017	0.0017	0.0002
2017	11	0.6193	0.0018	0.6213	0.0017	0.0020	0.0002
2017	12	0.6169	0.0018	0.6188	0.0017	0.0018	0.0003
2018	1	0.6158	0.0017	0.6176	0.0017	0.0018	0.0003
2018	2	0.6224	0.0018	0.6241	0.0018	0.0016	0.0003
2018	3	0.6213	0.0018	0.6231	0.0018	0.0018	0.0003
2018	4	0.6213	0.0018	0.6229	0.0018	0.0017	0.0003
2018	5	0.6235	0.0017	0.6253	0.0018	0.0017	0.0003
2018	6	0.6283	0.0015	0.6303	0.0015	0.0020	0.0003
2018	7	0.6300	0.0017	0.6318	0.0016	0.0018	0.0003
2018	8	0.6217	0.0018	0.6237	0.0018	0.0020	0.0003
2018	9	0.6210	0.0018	0.6230	0.0018	0.0020	0.0003
2018	10	0.6231	0.0016	0.6251	0.0016	0.0020	0.0002
2018	11	0.6237	0.0017	0.6258	0.0016	0.0021	0.0003
2018	12	0.6229	0.0017	0.6249	0.0017	0.0020	0.0003
2019	1	0.6220	0.0018	0.6239	0.0018	0.0019	0.0003
2019	2	0.6245	0.0018	0.6265	0.0017	0.0021	0.0003
2019	3	0.6227	0.0017	0.6252	0.0016	0.0025	0.0003
2019	4	0.6195	0.0017	0.6213	0.0017	0.0017	0.0003
2019	5	0.6213	0.0017	0.6236	0.0016	0.0023	0.0003
2019	6	0.6268	0.0017	0.6293	0.0017	0.0025	0.0003
2019	7	0.6312	0.0018	0.6340	0.0018	0.0028	0.0003
2019	8	0.6276	0.0015	0.6298	0.0015	0.0022	0.0003
2019	9	0.6272	0.0016	0.6295	0.0015	0.0022	0.0003
2019	10	0.6285	0.0017	0.6302	0.0016	0.0017	0.0003
2019	11	0.6266	0.0016	0.6287	0.0016	0.0021	0.0003
2019	12	0.6246	0.0016	0.6262	0.0016	0.0016	0.0003
2020	1	0.6233	0.0017	0.6248	0.0017	0.0016	0.0003

2020	2	0.6250	0.0018	0.6267	0.0017	0.0017	0.0003
2020	3	0.6183	0.0018	0.6195	0.0018	0.0013	0.0004
2020	4	0.5929	0.0018	0.5937	0.0019	0.0008	0.0005
2020	5	0.6002	0.0018	0.6017	0.0018	0.0015	0.0005
2020	6	0.6116	0.0016	0.6133	0.0016	0.0018	0.0005
2020	7	0.6150	0.0018	0.6179	0.0018	0.0029	0.0005
2020	8	0.6115	0.0018	0.6137	0.0019	0.0022	0.0005
2020	9	0.6087	0.0019	0.6107	0.0019	0.0020	0.0003
2020	10	0.6124	0.0017	0.6139	0.0018	0.0015	0.0004
2020	11	0.6085	0.0018	0.6105	0.0018	0.0020	0.0003
2020	12	0.6070	0.0019	0.6083	0.0018	0.0013	0.0003
2021	1	0.6039	0.0018	0.6050	0.0018	0.0011	0.0003
2021	2	0.6056	0.0018	0.6066	0.0018	0.0009	0.0004
2021	3	0.6065	0.0018	0.6074	0.0017	0.0009	0.0005
2021	4	0.6056	0.0020	0.6071	0.0019	0.0015	0.0004
2021	5	0.6070	0.0020	0.6090	0.0019	0.0019	0.0005
2021	6	0.6131	0.0019	0.6153	0.0019	0.0023	0.0004

Source: Basic Monthly CPS matched to administrative data. Table show how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells. CPS months used are January 2017 to June 2021.

Table A.3 Change in Unemployment Rate Estimates by State: January 2017 to February 2020

	Original		Adrec-CART		Difference	
State	Est.	S.E.	Est.	S.E.	Est.	S.E.
Alabama	0.0377	0.0017	0.0376	0.0017	-0.0001	0.0001
Alaska	0.0691	0.0056	0.0690	0.0056	-0.0001	0.0002
Arizona	0.0481	0.0019	0.0483	0.0019	0.0002	0.0002
Arkansas	0.0367	0.0018	0.0366	0.0018	-0.0001	0.0001
California	0.0437	0.0010	0.0437	0.0010	0.0000	0.0001
Colorado	0.0294	0.0016	0.0293	0.0015	-0.0001	0.0001
Connecticut	0.0440	0.0027	0.0437	0.0027	-0.0003	0.0002
Delaware	0.0432	0.0022	0.0432	0.0023	0.0000	0.0002
D.C.	0.0602	0.0023	0.0602	0.0023	-0.0001	0.0001
Florida	0.0366	0.0012	0.0365	0.0012	-0.0001	0.0001
Georgia	0.0408	0.0015	0.0408	0.0015	0.0000	0.0001
Hawaii	0.0273	0.0014	0.0274	0.0014	0.0001	0.0001
Idaho	0.0324	0.0014	0.0324	0.0014	0.0001	0.0001
Illinois	0.0439	0.0018	0.0441	0.0018	0.0002	0.0001
Indiana	0.0365	0.0020	0.0366	0.0020	0.0000	0.0001
Iowa	0.0293	0.0029	0.0292	0.0029	-0.0001	0.0001
Kansas	0.0337	0.0017	0.0337	0.0017	0.0000	0.0001
Kentucky	0.0459	0.0034	0.0459	0.0034	0.0000	0.0002

Louisiana	0.0493	0.0026	0.0495	0.0026	0.0002	0.0002
Maine	0.0345	0.0026	0.0345	0.0025	0.0000	0.0001
Maryland	0.0403	0.0018	0.0401	0.0018	-0.0002	0.0001
Massachusetts	0.0348	0.0013	0.0350	0.0014	0.0002	0.0001
Michigan	0.0425	0.0020	0.0424	0.0020	-0.0002	0.0001
Minnesota	0.0325	0.0014	0.0325	0.0014	0.0001	0.0001
Mississippi	0.0524	0.0023	0.0522	0.0023	-0.0001	0.0001
Missouri	0.0352	0.0021	0.0352	0.0021	0.0001	0.0001
Montana	0.0384	0.0018	0.0383	0.0018	-0.0001	0.0001
Nebraska	0.0307	0.0016	0.0306	0.0016	-0.0001	0.0001
Nevada	0.0452	0.0019	0.0451	0.0019	-0.0001	0.0001
New Hampshire	0.0262	0.0013	0.0261	0.0014	-0.0001	0.0001
New Jersey	0.0421	0.0016	0.0418	0.0016	-0.0003	0.0001
New Mexico	0.0530	0.0029	0.0530	0.0029	0.0000	0.0001
New York	0.0426	0.0011	0.0426	0.0011	0.0000	0.0001
North Carolina	0.0414	0.0021	0.0415	0.0021	0.0000	0.0001
North Dakota	0.0258	0.0016	0.0259	0.0016	0.0001	0.0001
Ohio	0.0463	0.0015	0.0461	0.0015	-0.0002	0.0002
Oklahoma	0.0385	0.0022	0.0384	0.0023	0.0000	0.0002
Oregon	0.0412	0.0017	0.0411	0.0017	-0.0001	0.0001
Pennsylvania	0.0446	0.0014	0.0445	0.0014	-0.0001	0.0001
Rhode Island	0.0410	0.0023	0.0409	0.0023	-0.0001	0.0001
South Carolina	0.0370	0.0023	0.0368	0.0023	-0.0002	0.0001
South Dakota	0.0340	0.0029	0.0340	0.0028	0.0000	0.0002
Tennessee	0.0359	0.0014	0.0361	0.0014	0.0002	0.0001
Texas	0.0394	0.0011	0.0395	0.0011	0.0001	0.0001
Utah	0.0319	0.0021	0.0320	0.0022	0.0000	0.0001
Vermont	0.0279	0.0015	0.0278	0.0015	-0.0001	0.0001
Virginia	0.0322	0.0014	0.0320	0.0014	-0.0002	0.0001
Washington	0.0441	0.0024	0.0444	0.0024	0.0002	0.0001
West Virginia	0.0521	0.0031	0.0520	0.0030	-0.0002	0.0001
Wisconsin	0.0328	0.0014	0.0327	0.0013	0.0000	0.0001
Wyoming	0.0402	0.0021	0.0404	0.0021	0.0001	0.0001

Source: Basic Monthly CPS matched to administrative data. Table shows by state how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells.

Table A.4 Change in Unemployment Rate Estimates by State: April to August 2020

	Original		Adrec-CART		Difference	
State	Est.	S.E.	Est.	S.E.	Est.	S.E.
Alabama	0.0855	0.0069	0.0863	0.0070	0.0008	0.0007
Alaska	0.0987	0.0115	0.0984	0.0117	-0.0003	0.0011
Arizona	0.1035	0.0066	0.1036	0.0067	0.0000	0.0010

Arkansas	0.0832	0.0071	0.0841	0.0070	0.0008	0.0010
California	0.1428	0.0042	0.1449	0.0043	0.0021	0.0006
Colorado	0.0996	0.0075	0.1020	0.0076	0.0024	0.0010
Connecticut	0.1012	0.0083	0.1032	0.0086	0.0021	0.0013
Delaware	0.1198	0.0093	0.1233	0.0097	0.0035	0.0014
D.C.	0.0943	0.0073	0.0969	0.0074	0.0026	0.0010
Florida	0.1186	0.0058	0.1189	0.0059	0.0003	0.0009
Georgia	0.0868	0.0072	0.0868	0.0070	0.0001	0.0010
Hawaii	0.1629	0.0113	0.1659	0.0118	0.0029	0.0017
Idaho	0.0722	0.0062	0.0726	0.0062	0.0004	0.0006
Illinois	0.1346	0.0068	0.1359	0.0069	0.0013	0.0010
Indiana	0.1076	0.0067	0.1090	0.0068	0.0015	0.0012
Iowa	0.0801	0.0093	0.0813	0.0094	0.0013	0.0009
Kansas	0.0779	0.0068	0.0793	0.0071	0.0014	0.0010
Kentucky	0.0967	0.0101	0.1018	0.0106	0.0052	0.0013
Louisiana	0.1123	0.0067	0.1141	0.0070	0.0018	0.0010
Maine	0.0716	0.0069	0.0717	0.0070	0.0000	0.0010
Maryland	0.0872	0.0073	0.0875	0.0073	0.0003	0.0008
Massachusetts	0.1488	0.0091	0.1531	0.0093	0.0044	0.0012
Michigan	0.1568	0.0083	0.1565	0.0082	-0.0003	0.0010
Minnesota	0.0842	0.0065	0.0857	0.0064	0.0015	0.0009
Mississippi	0.1083	0.0093	0.1108	0.0104	0.0025	0.0016
Missouri	0.0832	0.0087	0.0853	0.0092	0.0021	0.0011
Montana	0.0782	0.0087	0.0814	0.0090	0.0032	0.0009
Nebraska	0.0602	0.0063	0.0610	0.0063	0.0008	0.0008
Nevada	0.2077	0.0140	0.2099	0.0145	0.0023	0.0024
New Hampshire	0.1027	0.0078	0.1043	0.0080	0.0017	0.0012
New Jersey	0.1458	0.0088	0.1481	0.0090	0.0023	0.0011
New Mexico	0.1087	0.0077	0.1110	0.0083	0.0023	0.0014
New York	0.1469	0.0058	0.1470	0.0059	0.0000	0.0010
North Carolina	0.1009	0.0058	0.1027	0.0061	0.0018	0.0013
North Dakota	0.0705	0.0086	0.0713	0.0087	0.0008	0.0008
Ohio	0.1249	0.0066	0.1256	0.0066	0.0008	0.0012
Oklahoma	0.0826	0.0096	0.0816	0.0100	-0.0010	0.0012
Oregon	0.1056	0.0103	0.1089	0.0102	0.0034	0.0010
Pennsylvania	0.1326	0.0065	0.1340	0.0064	0.0014	0.0012
Rhode Island	0.1352	0.0123	0.1375	0.0128	0.0023	0.0014
South Carolina	0.0884	0.0073	0.0886	0.0073	0.0002	0.0010
South Dakota	0.0652	0.0117	0.0668	0.0128	0.0016	0.0017
Tennessee	0.1114	0.0079	0.1142	0.0082	0.0028	0.0010
Texas	0.1004	0.0046	0.1010	0.0046	0.0006	0.0006
Utah	0.0701	0.0053	0.0718	0.0056	0.0018	0.0007
Vermont	0.0913	0.0069	0.0925	0.0069	0.0012	0.0010
Virginia	0.0901	0.0062	0.0918	0.0063	0.0017	0.0009

Washington	0.1161	0.0078	0.1168	0.0075	0.0006	0.0013
West Virginia	0.1106	0.0058	0.1126	0.0055	0.0020	0.0009
Wisconsin	0.0912	0.0062	0.0918	0.0063	0.0007	0.0013
Wyoming	0.0677	0.0051	0.0689	0.0052	0.0012	0.0010

Source: Basic Monthly CPS matched to administrative data. Table shows by state how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells.

Table A.5 Change in Labor Force Particpation Rate Estimates by State: January 2017 to February 2020

	Original		Adrec-CART		Difference	
State	Est.	S.E.	Est.	S.E.	Est.	S.E.
Alabama	0.5694	0.0058	0.5715	0.0059	0.0022	0.0002
Alaska	0.6428	0.0063	0.6440	0.0063	0.0012	0.0003
Arizona	0.6054	0.0092	0.6073	0.0092	0.0019	0.0003
Arkansas	0.5738	0.0083	0.5761	0.0086	0.0023	0.0004
California	0.6172	0.0024	0.6189	0.0024	0.0017	0.0001
Colorado	0.6834	0.0077	0.6855	0.0079	0.0021	0.0004
Connecticut	0.6552	0.0064	0.6576	0.0063	0.0024	0.0004
Delaware	0.6224	0.0055	0.6257	0.0054	0.0033	0.0003
D.C.	0.7021	0.0038	0.7038	0.0038	0.0017	0.0002
Florida	0.5893	0.0036	0.5916	0.0036	0.0022	0.0003
Georgia	0.6205	0.0058	0.6219	0.0060	0.0014	0.0004
Hawaii	0.6113	0.0050	0.6117	0.0050	0.0003	0.0003
Idaho	0.6312	0.0061	0.6336	0.0061	0.0024	0.0003
Illinois	0.6390	0.0042	0.6419	0.0042	0.0029	0.0003
Indiana	0.6384	0.0062	0.6412	0.0061	0.0028	0.0003
Iowa	0.6857	0.0040	0.6887	0.0039	0.0030	0.0003
Kansas	0.6603	0.0055	0.6627	0.0055	0.0024	0.0003
Kentucky	0.5863	0.0105	0.5858	0.0109	-0.0004	0.0008
Louisiana	0.5825	0.0044	0.5837	0.0044	0.0012	0.0004
Maine	0.6224	0.0090	0.6242	0.0090	0.0018	0.0004
Maryland	0.6774	0.0052	0.6795	0.0053	0.0021	0.0005
Massachusetts	0.6659	0.0041	0.6673	0.0041	0.0013	0.0002
Michigan	0.6095	0.0045	0.6126	0.0045	0.0032	0.0003
Minnesota	0.6973	0.0062	0.6990	0.0061	0.0017	0.0004
Mississippi	0.5474	0.0044	0.5491	0.0045	0.0018	0.0005
Missouri	0.6283	0.0067	0.6297	0.0068	0.0014	0.0004
Montana	0.6247	0.0060	0.6265	0.0061	0.0019	0.0002
Nebraska	0.6894	0.0043	0.6919	0.0043	0.0024	0.0004
Nevada	0.6233	0.0059	0.6257	0.0058	0.0024	0.0003
New Hampshire	0.6809	0.0055	0.6835	0.0055	0.0026	0.0003

New Jersey	0.6221	0.0041	0.6243	0.0041	0.0023	0.0003
New Mexico	0.5746	0.0067	0.5769	0.0066	0.0023	0.0004
New York	0.6023	0.0032	0.6040	0.0032	0.0017	0.0002
North Carolina	0.6067	0.0049	0.6095	0.0050	0.0028	0.0003
North Dakota	0.6911	0.0079	0.6929	0.0080	0.0018	0.0003
Ohio	0.6236	0.0046	0.6255	0.0045	0.0019	0.0003
Oklahoma	0.6042	0.0097	0.6058	0.0099	0.0015	0.0004
Oregon	0.6208	0.0073	0.6236	0.0072	0.0028	0.0004
Pennsylvania	0.6246	0.0046	0.6272	0.0046	0.0026	0.0004
Rhode Island	0.6448	0.0057	0.6469	0.0057	0.0021	0.0003
South Carolina	0.5797	0.0052	0.5823	0.0051	0.0026	0.0003
South Dakota	0.6843	0.0080	0.6860	0.0081	0.0016	0.0004
Tennessee	0.6068	0.0045	0.6078	0.0045	0.0010	0.0005
Texas	0.6313	0.0027	0.6326	0.0028	0.0013	0.0002
Utah	0.6749	0.0071	0.6766	0.0072	0.0017	0.0003
Vermont	0.6584	0.0053	0.6608	0.0052	0.0024	0.0003
Virginia	0.6489	0.0052	0.6512	0.0053	0.0023	0.0004
Washington	0.6370	0.0087	0.6377	0.0091	0.0007	0.0006
West Virginia	0.5376	0.0121	0.5386	0.0125	0.0009	0.0005
Wisconsin	0.6705	0.0048	0.6731	0.0048	0.0026	0.0003
Wyoming	0.6444	0.0069	0.6460	0.0069	0.0016	0.0003

Source: Basic Monthly CPS matched to administrative data. Table shows by state how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells.

Table A.6 Change in Labor Force Participation Rate Estimates by State: April to August 2020

	Original		Adrec-CART		Difference	
State	Est.	S.E.	Est.	S.E.	Est.	S.E.
Alabama	0.5802	0.0091	0.5799	0.0093	-0.0003	0.0011
Alaska	0.6159	0.0124	0.6152	0.0129	-0.0007	0.0020
Arizona	0.6097	0.0132	0.6101	0.0127	0.0005	0.0019
Arkansas	0.5689	0.0113	0.5733	0.0115	0.0044	0.0022
California	0.5927	0.0046	0.5937	0.0045	0.0010	0.0007
Colorado	0.6576	0.0136	0.6612	0.0136	0.0036	0.0013
Connecticut	0.6300	0.0123	0.6308	0.0123	0.0008	0.0018
Delaware	0.6021	0.0141	0.6068	0.0142	0.0047	0.0014
D.C.	0.6800	0.0104	0.6805	0.0104	0.0005	0.0014
Florida	0.5510	0.0064	0.5532	0.0064	0.0021	0.0011
Georgia	0.5928	0.0098	0.5930	0.0104	0.0002	0.0019
Hawaii	0.5800	0.0126	0.5793	0.0131	-0.0007	0.0021
Idaho	0.6341	0.0128	0.6354	0.0123	0.0012	0.0018
Illinois	0.6151	0.0080	0.6168	0.0079	0.0017	0.0012
Indiana	0.6145	0.0105	0.6171	0.0099	0.0026	0.0015

Iowa	0.6588	0.0104	0.6616	0.0101	0.0028	0.0016
Kansas	0.6774	0.0125	0.6818	0.0119	0.0044	0.0015
Kentucky	0.5631	0.0136	0.5586	0.0141	-0.0045	0.0026
Louisiana	0.5604	0.0079	0.5625	0.0081	0.0020	0.0014
Maine	0.6053	0.0184	0.6048	0.0184	-0.0005	0.0020
Maryland	0.6699	0.0105	0.6700	0.0105	0.0001	0.0013
Massachusetts	0.6328	0.0104	0.6328	0.0105	0.0000	0.0013
Michigan	0.5946	0.0083	0.6004	0.0084	0.0058	0.0012
Minnesota	0.6837	0.0107	0.6867	0.0110	0.0029	0.0015
Mississippi	0.5399	0.0109	0.5408	0.0113	0.0009	0.0014
Missouri	0.6270	0.0085	0.6304	0.0086	0.0034	0.0012
Montana	0.6312	0.0102	0.6331	0.0101	0.0020	0.0014
Nebraska	0.6969	0.0143	0.6969	0.0138	0.0000	0.0013
Nevada	0.6055	0.0110	0.6106	0.0113	0.0052	0.0018
New Hampshire	0.6596	0.0108	0.6602	0.0112	0.0006	0.0018
New Jersey	0.6359	0.0093	0.6370	0.0096	0.0011	0.0014
New Mexico	0.5579	0.0099	0.5622	0.0100	0.0043	0.0023
New York	0.5767	0.0068	0.5771	0.0067	0.0004	0.0011
North Carolina	0.5745	0.0112	0.5793	0.0113	0.0048	0.0017
North Dakota	0.6904	0.0111	0.6896	0.0118	-0.0007	0.0016
Ohio	0.6034	0.0084	0.6055	0.0084	0.0021	0.0014
Oklahoma	0.6034	0.0143	0.6022	0.0147	-0.0012	0.0020
Oregon	0.6177	0.0115	0.6199	0.0108	0.0022	0.0018
Pennsylvania	0.6125	0.0084	0.6146	0.0083	0.0022	0.0015
Rhode Island	0.6107	0.0134	0.6110	0.0134	0.0002	0.0021
South Carolina	0.5693	0.0129	0.5718	0.0125	0.0026	0.0016
South Dakota	0.6988	0.0107	0.6966	0.0111	-0.0022	0.0020
Tennessee	0.5871	0.0085	0.5913	0.0087	0.0042	0.0016
Texas	0.6145	0.0056	0.6157	0.0058	0.0012	0.0009
Utah	0.6662	0.0096	0.6699	0.0095	0.0037	0.0011
Vermont	0.6378	0.0099	0.6371	0.0099	-0.0008	0.0018
Virginia	0.6547	0.0111	0.6594	0.0112	0.0047	0.0017
Washington	0.6365	0.0088	0.6407	0.0088	0.0042	0.0013
West Virginia	0.5355	0.0100	0.5351	0.0102	-0.0003	0.0015
Wisconsin	0.6586	0.0156	0.6604	0.0153	0.0018	0.0015
Wyoming	0.6467	0.0105	0.6476	0.0096	0.0008	0.0021

Source: Basic Monthly CPS matched to administrative data. Table shows by state how CPS estimates change with using new weights based on administrative data and using CART to create new noninterview adjustment cells.

Table A.7 Comparison of Administrative Data-Based Weighting for the CPS

		Unemployment			Labor Force Participation		
Year	Month	Original	Adrec-CART	Adrec-EBW	Original	Adrec-CART	Adrec-EBW
2017	1	5.17	5.17	5.18	61.74	61.90	61.82
2017	2	4.98	4.98	5.01	62.07	62.21	62.13
2017	3	4.61	4.60	4.62	62.12	62.31	62.24
2017	4	4.17	4.16	4.20	62.22	62.39	62.35
2017	5	4.15	4.15	4.15	62.17	62.35	62.33
2017	6	4.57	4.55	4.53	62.63	62.80	62.76
2017	7	4.68	4.66	4.67	62.96	63.14	63.10
2017	8	4.62	4.60	4.59	62.36	62.56	62.50
2017	9	4.10	4.09	4.10	62.37	62.55	62.48
2017	10	3.95	3.94	3.95	61.98	62.15	62.16
2017	11	3.95	3.94	3.96	61.93	62.13	62.06
2017	12	3.96	3.97	3.97	61.69	61.88	61.82
2018	1	4.57	4.56	4.57	61.58	61.76	61.72
2018	2	4.44	4.44	4.46	62.24	62.41	62.28
2018	3	4.17	4.18	4.18	62.13	62.31	62.24
2018	4	3.74	3.76	3.74	62.13	62.29	62.28
2018	5	3.61	3.61	3.62	62.35	62.53	62.51
2018	6	4.27	4.27	4.28	62.83	63.03	63.04
2018	7	4.14	4.14	4.14	63.00	63.18	63.15
2018	8	4.03	4.03	4.00	62.17	62.37	62.36
2018	9	3.62	3.61	3.63	62.10	62.30	62.27
2018	10	3.62	3.60	3.62	62.31	62.51	62.45
2018	11	3.53	3.52	3.54	62.37	62.58	62.53
2018	12	3.74	3.74	3.75	62.29	62.49	62.45
2019	1	4.49	4.51	4.49	62.20	62.39	62.34
2019	2	4.16	4.15	4.13	62.45	62.65	62.61
2019	3	3.97	3.98	3.97	62.27	62.52	62.48
2019	4	3.38	3.39	3.39	61.95	62.13	62.05
2019	5	3.42	3.43	3.44	62.13	62.36	62.30
2019	6	3.92	3.92	3.93	62.68	62.93	62.89
2019	7	4.07	4.06	4.07	63.12	63.40	63.32
2019	8	3.87	3.87	3.86	62.76	62.98	62.88
2019	9	3.38	3.38	3.39	62.72	62.95	62.84
2019	10	3.37	3.38	3.37	62.85	63.02	62.87
2019	11	3.34	3.35	3.34	62.66	62.87	62.73
2019	12	3.41	3.42	3.43	62.46	62.62	62.51
2020	1	4.05	4.07	4.05	62.33	62.48	62.44
2020	2	3.87	3.88	3.89	62.50	62.67	62.63
2020	3	4.62	4.67	4.72	61.83	61.95	61.90
2020	4	14.56	14.78	14.85	59.29	59.37	59.18
2020	5	13.04	13.24	13.35	60.02	60.17	59.92
2020	6	11.24	11.36	11.64	61.16	61.33	61.01

2020	7	10.41	10.49	10.68	61.50	61.79	61.39
2020	8	8.47	8.50	8.73	61.15	61.37	61.01
2020	9	7.63	7.69	7.77	60.87	61.07	60.88
2020	10	6.57	6.60	6.65	61.24	61.39	61.25
2020	11	6.36	6.39	6.45	60.85	61.05	60.94
2020	12	6.50	6.57	6.64	60.70	60.83	60.68
2021	1	6.87	6.98	7.00	60.39	60.50	60.41
2021	2	6.67	6.73	6.79	60.56	60.66	60.53
2021	3	6.23	6.31	6.37	60.65	60.74	60.66
2021	4	5.82	5.87	5.90	60.56	60.71	60.60
2021	5	5.66	5.68	5.71	60.70	60.90	60.79
2021	6	6.23	6.23	6.27	61.31	61.53	61.42

		College Graduate (Age 25 and Over)			Married		
Year	Month	Original	Adrec-CART	Adrec-EBW	Original	Adrec-CART	Adrec-EBW
2017	1	34.10	34.02	34.14	50.73	50.58	50.29
2017	2	34.54	34.46	34.59	50.78	50.61	50.26
2017	3	34.25	34.20	34.26	50.81	50.67	50.42
2017	4	34.25	34.19	34.34	50.73	50.57	50.35
2017	5	34.47	34.39	34.49	50.85	50.72	50.42
2017	6	34.52	34.45	34.56	50.89	50.73	50.47
2017	7	34.86	34.77	34.94	50.72	50.55	50.23
2017	8	34.65	34.60	34.73	50.51	50.36	50.02
2017	9	34.52	34.42	34.67	50.40	50.27	50.04
2017	10	34.64	34.58	34.71	50.49	50.38	50.13
2017	11	34.62	34.54	34.65	50.33	50.24	49.86
2017	12	34.87	34.76	34.89	50.32	50.20	49.86
2018	1	34.87	34.83	34.95	50.42	50.30	49.94
2018	2	35.18	35.10	35.24	50.23	50.10	49.66
2018	3	34.97	34.88	35.03	50.31	50.21	49.89
2018	4	34.97	34.87	35.01	50.40	50.27	50.00
2018	5	34.98	34.85	35.03	50.27	50.12	49.91
2018	6	34.90	34.77	34.90	50.14	49.96	49.80
2018	7	35.32	35.21	35.36	49.93	49.78	49.56
2018	8	35.42	35.30	35.42	50.10	49.98	49.69
2018	9	35.34	35.25	35.32	50.13	50.04	49.71
2018	10	35.77	35.68	35.69	50.04	49.99	49.62
2018	11	35.95	35.79	35.79	50.26	50.13	49.69
2018	12	36.17	35.96	35.97	50.26	50.07	49.68
2019	1	35.89	35.73	35.78	50.29	50.09	49.72
2019	2	36.09	35.95	36.06	50.11	49.97	49.61
2019	3	36.02	35.83	35.89	50.21	50.00	49.72
2019	4	35.80	35.58	35.61	49.95	49.72	49.44

2019	5	35.71	35.50	35.66	49.87	49.64	49.40
2019	6	35.82	35.65	35.71	49.72	49.58	49.24
2019	7	36.11	35.98	36.00	49.65	49.53	49.06
2019	8	35.94	35.74	35.83	49.98	49.81	49.32
2019	9	36.07	35.97	35.94	49.84	49.70	49.13
2019	10	36.38	36.22	36.22	49.79	49.63	49.00
2019	11	36.52	36.32	36.32	49.72	49.56	48.91
2019	12	36.70	36.42	36.47	49.73	49.56	48.93
2020	1	36.79	36.59	36.55	50.17	49.93	49.52
2020	2	36.89	36.75	36.66	50.11	49.97	49.63
2020	3	37.30	36.95	36.67	50.62	50.23	49.67
2020	4	37.77	37.30	36.85	50.79	50.37	49.41
2020	5	37.77	37.30	36.65	51.11	50.62	49.32
2020	6	38.21	37.85	36.93	50.93	50.41	48.97
2020	7	38.52	38.27	37.30	50.78	50.32	48.89
2020	8	38.17	37.91	37.05	50.77	50.35	49.03
2020	9	37.19	36.95	36.62	49.89	49.62	48.90
2020	10	37.06	36.84	36.51	49.87	49.61	48.97
2020	11	37.00	36.80	36.53	49.62	49.41	48.85
2020	12	37.34	37.07	36.61	49.57	49.36	48.61
2021	1	37.47	37.14	36.80	49.65	49.38	48.79
2021	2	37.83	37.48	37.19	49.76	49.49	48.83
2021	3	37.78	37.39	37.18	49.58	49.28	48.64
2021	4	37.64	37.31	37.19	49.63	49.31	48.65
2021	5	37.42	37.14	37.06	49.55	49.20	48.60
2021	6	37.31	37.05	36.94	49.31	49.05	48.53

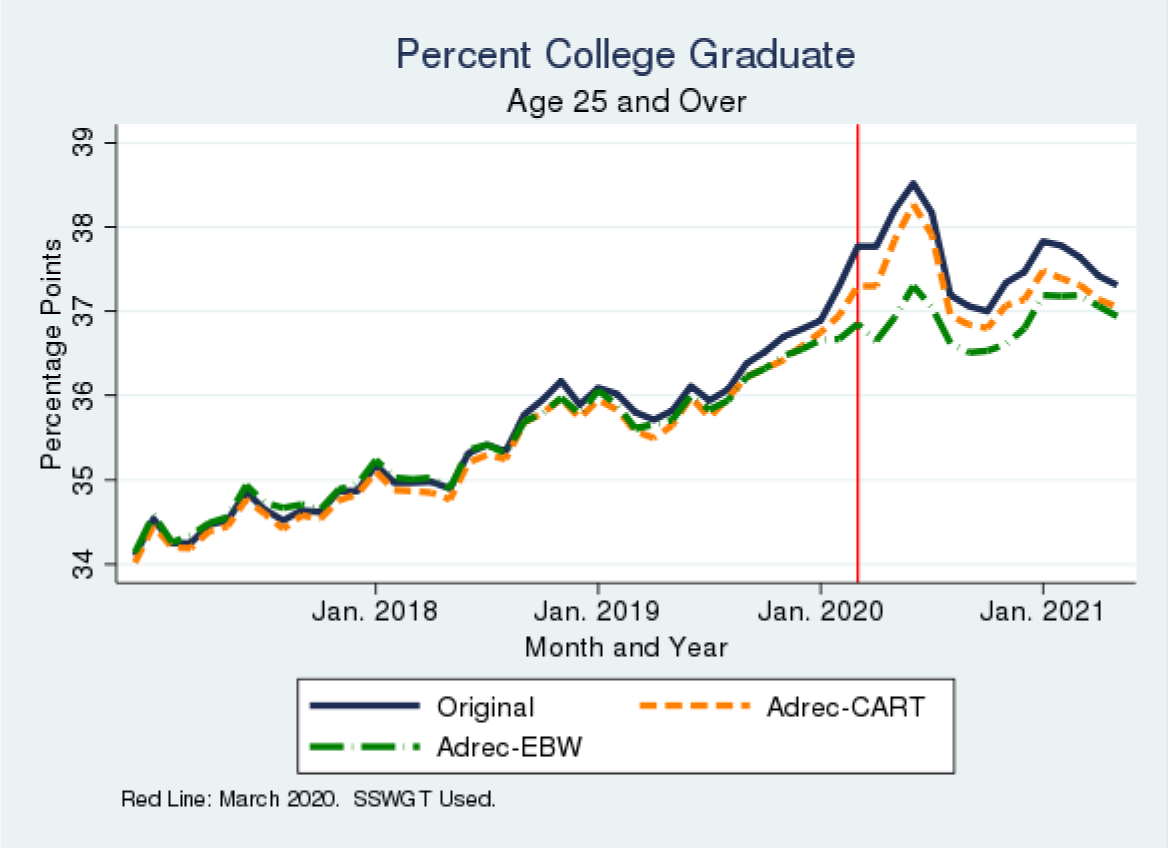
		Unemployment			Labor Force Participation		
Year	Month	Original	Adrec-CART	Adrec-EBW	Original	Adrec-CART	Adrec-EBW
2017	1	83.88	83.87	84.13	8.43	8.45	8.28
2017	2	83.97	83.97	84.29	8.30	8.33	8.12
2017	3	83.96	83.97	84.22	8.36	8.37	8.22
2017	4	83.85	83.82	84.20	8.36	8.40	8.18
2017	5	83.93	83.91	84.33	8.21	8.25	8.01
2017	6	83.87	83.84	84.19	8.40	8.45	8.25
2017	7	83.93	83.93	84.31	8.27	8.31	8.08
2017	8	83.88	83.89	84.24	8.31	8.34	8.14
2017	9	83.90	83.91	84.19	8.19	8.22	8.05
2017	10	83.91	83.92	84.25	8.18	8.21	8.04
2017	11	83.84	83.83	84.16	8.08	8.12	7.98
2017	12	83.95	83.92	84.24	8.07	8.14	7.96
2018	1	83.57	83.55	83.88	8.40	8.44	8.24
2018	2	83.47	83.48	83.82	8.36	8.38	8.18

2018	3	83.33	83.33	83.66	8.53	8.56	8.35
2018	4	83.43	83.44	83.76	8.31	8.34	8.16
2018	5	83.60	83.61	83.96	8.19	8.23	8.04
2018	6	83.67	83.67	84.01	8.17	8.23	8.04
2018	7	83.78	83.77	84.18	8.07	8.12	7.91
2018	8	83.64	83.60	83.96	8.22	8.28	8.08
2018	9	83.49	83.48	83.84	8.20	8.23	8.05
2018	10	83.52	83.52	83.86	8.21	8.25	8.07
2018	11	83.45	83.43	83.82	8.16	8.21	8.01
2018	12	83.57	83.57	83.88	8.16	8.21	8.04
2019	1	83.55	83.57	83.84	8.33	8.37	8.22
2019	2	83.36	83.41	83.69	8.34	8.35	8.19
2019	3	83.32	83.39	83.65	8.38	8.40	8.23
2019	4	83.35	83.40	83.64	8.28	8.31	8.17
2019	5	83.41	83.46	83.77	8.13	8.15	7.96
2019	6	83.50	83.52	83.77	7.93	7.96	7.84
2019	7	83.68	83.68	83.97	7.73	7.79	7.65
2019	8	83.67	83.66	83.90	7.66	7.71	7.63
2019	9	83.72	83.70	83.91	7.72	7.79	7.71
2019	10	83.89	83.86	84.13	7.74	7.79	7.68
2019	11	83.69	83.69	84.00	7.86	7.90	7.75
2019	12	83.87	83.84	84.09	7.89	7.96	7.82
2020	1	83.85	83.82	84.01	7.86	7.93	7.83
2020	2	83.64	83.66	83.83	7.84	7.89	7.78
2020	3	83.90	83.79	83.93	7.76	7.90	7.83
2020	4	83.85	83.70	84.04	7.53	7.72	7.60
2020	5	83.85	83.68	84.09	7.44	7.64	7.50
2020	6	83.96	83.81	84.22	7.14	7.34	7.23
2020	7	84.02	83.91	84.18	7.21	7.34	7.33
2020	8	84.06	83.91	84.29	7.22	7.40	7.30
2020	9	84.14	84.13	84.35	7.38	7.44	7.38
2020	10	84.12	84.09	84.26	7.71	7.77	7.69
2020	11	83.82	83.79	84.02	7.98	8.04	7.92
2020	12	83.75	83.67	83.96	8.00	8.11	7.97
2021	1	83.68	83.58	83.84	7.85	7.96	7.83
2021	2	83.56	83.52	83.79	7.87	7.94	7.80
2021	3	83.68	83.60	83.89	7.80	7.90	7.72
2021	4	83.68	83.63	83.96	7.73	7.80	7.61
2021	5	83.77	83.72	84.06	7.71	7.79	7.59
2021	6	83.79	83.76	84.12	7.55	7.63	7.42

Table compares estimates from Basic Monthly CPS using various weighting methodology. “Original” is the second stage weights from the existing production method. “Adrec-CART” is the administrative data method as described in Section 4 which used CART to create new cells for the household noninterview adjustment, leaving the rest of weighting algorithm the same. “Adrec-EBW” is the administrative data method described in Section 6 which use entropy-balancing to create weights, replace all steps

of the existing CPS weighting algorithm, including the second-stage steps which calibrate to estimates from Census's Population Estimate Program. The estimate for percent that are college graduates only includes individuals over 25. All other estimates are conditional on the respondent being over 15.
Source: Basic Monthly CPS matched to administrative data

Figure A.3 Change in Percent College Graduate Estimates from Administrative-Based Weighting Methods



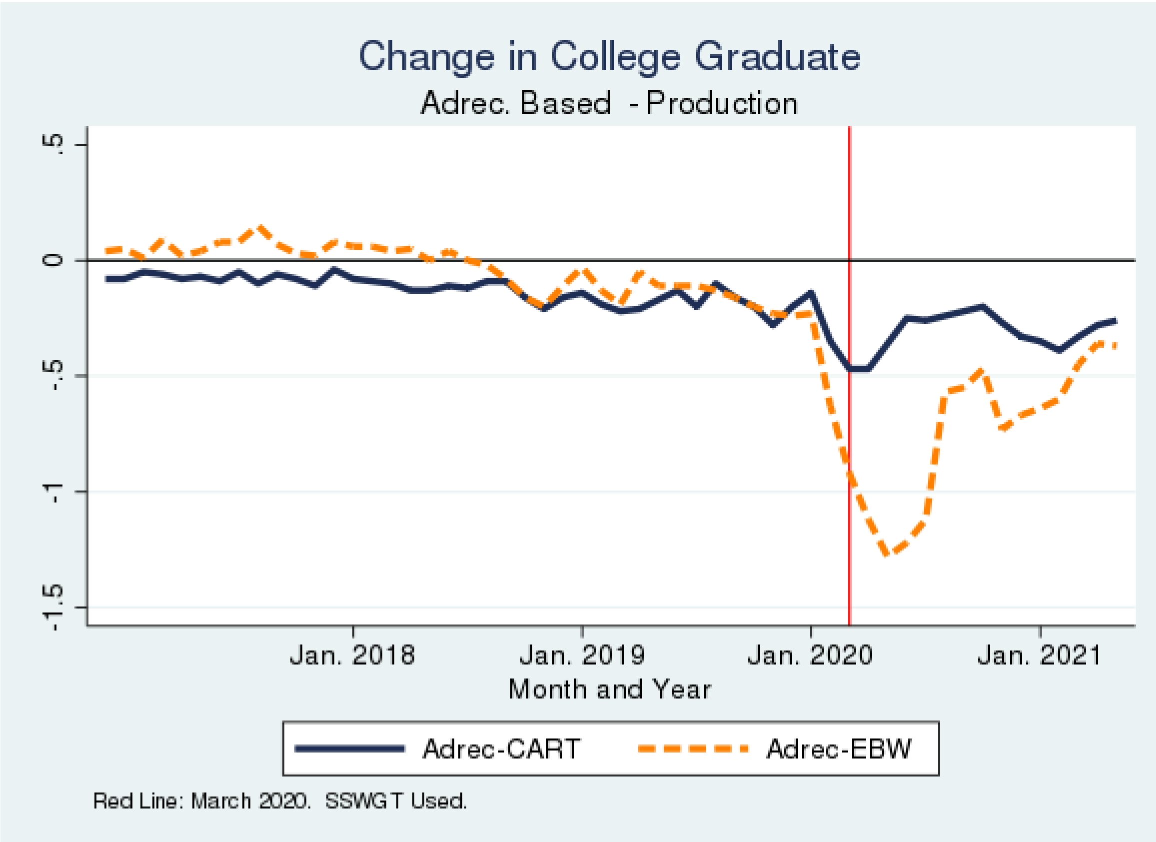
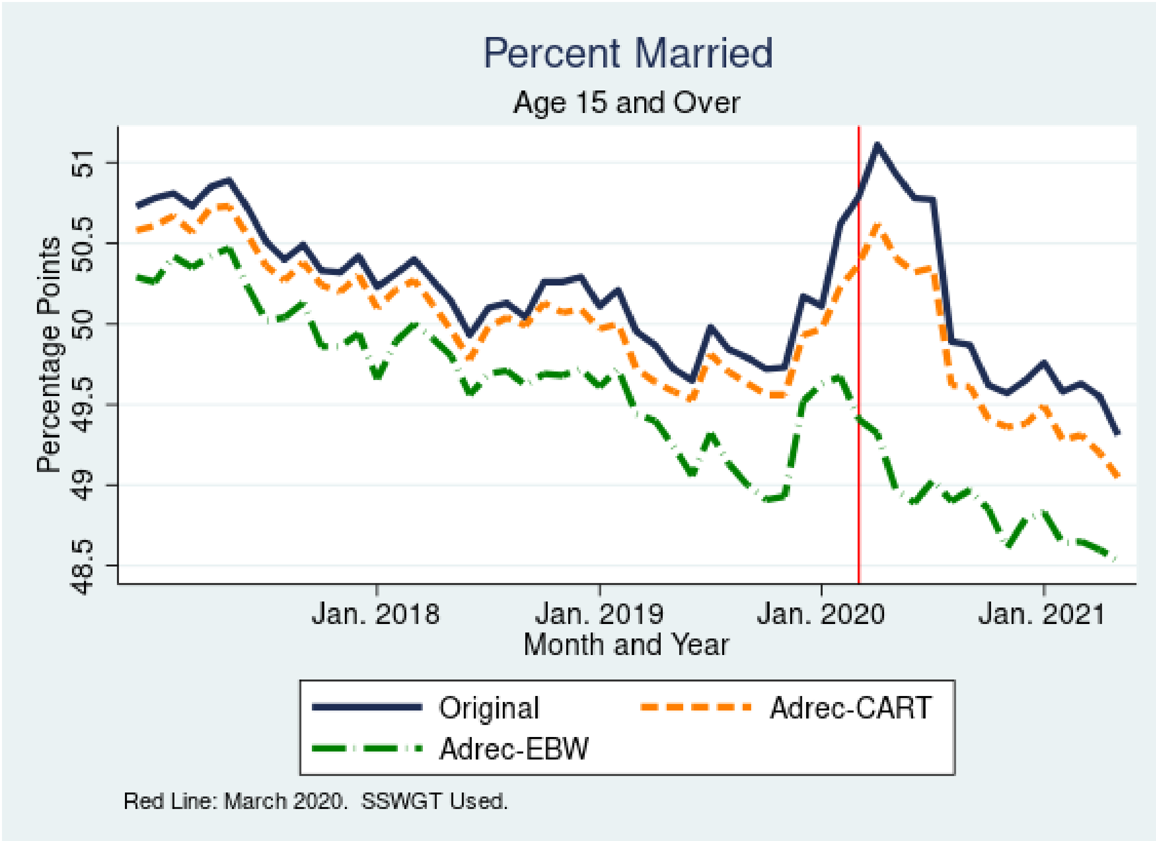


Figure present both levels (top panel) and the difference between the production estimate and the administrative data-based estimate (bottom panel). See Table 8 for more details.

Source: Basic Monthly CPS matched to administrative data

Figure A.4 Change in Percent Married Estimates from Administrative-Based Weighting Methods



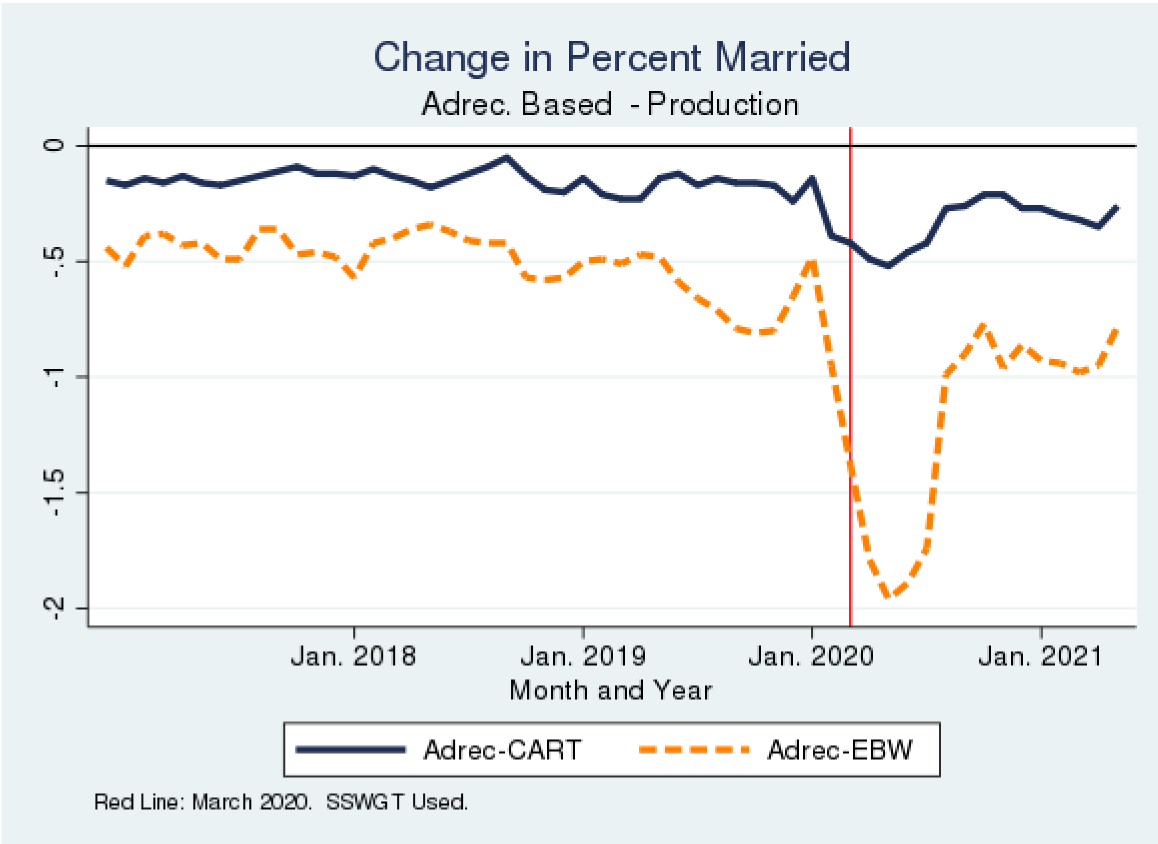
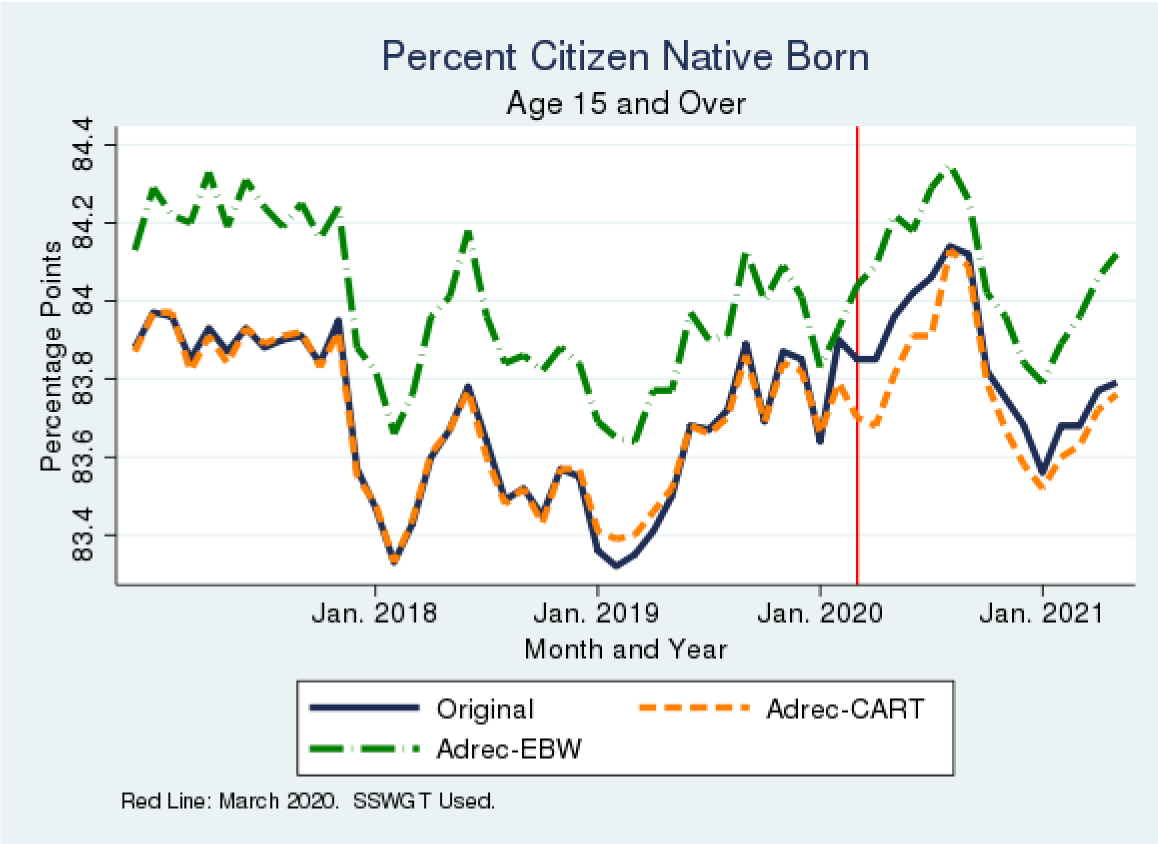


Figure present both levels (top panel) and the difference between the production estimate and the administrative –data-based estimate (bottom panel). See Table 8 for more details.

Source: Basic Monthly CPS matched to administrative data

Figure A.5 Change in Percent Citizen Native Born Estimates from Administrative-Based Weighting Method


Percent Citizen Native Born
Age 15 and Over
84.4
84.2
84
83.8
83.6
83.4
Percentage Points
Jan. 2018
Jan. 2019
Jan. 2020
Jan. 2021
Month and Year
Original
Adrec-CART
Adrec-EBW
Red Line: March 2020. SSWGT Used.

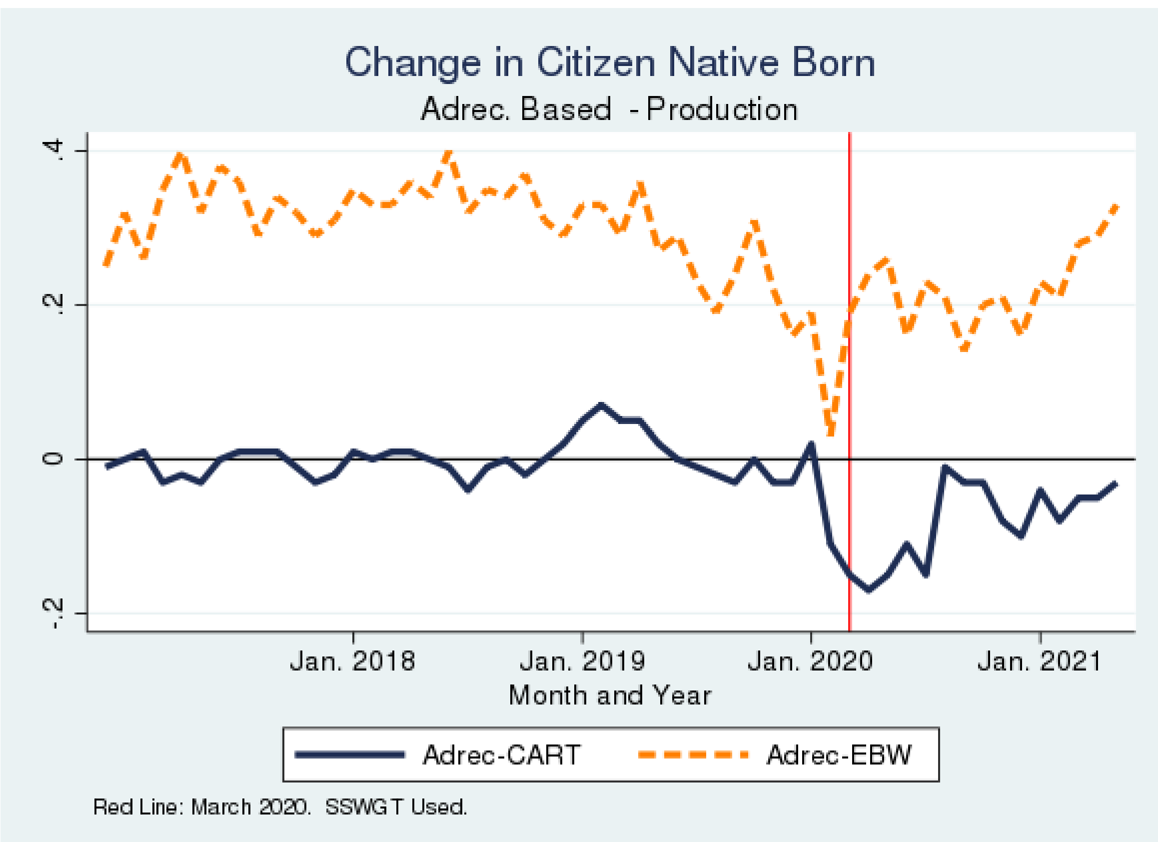
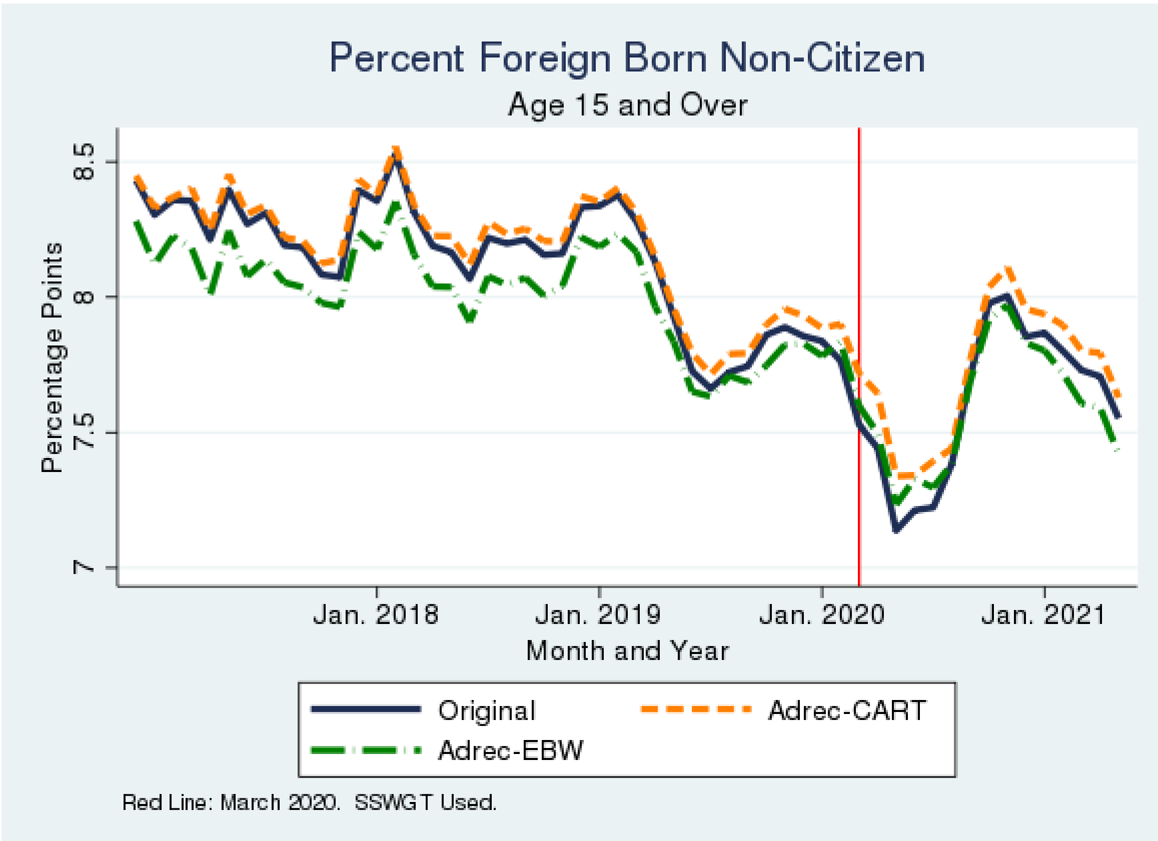

Change in Citizen Native Born
Adrec. Based - Production
.4
.2
0
-.2
Jan. 2018
Jan. 2019
Jan. 2020
Jan. 2021
Month and Year
Adrec-CART
Adrec-EBW
Red Line: March 2020. SSWGT Used.

Figure present both levels (top panel) and the difference between the production estimate and the administrative-data based estimate (bottom panel). See Table 8 for more details.

Source: Basic Monthly CPS matched to administrative data

Figure A.6 Change in Percent Non-Citizen Estimates from Administrative-Based Weighting Methods



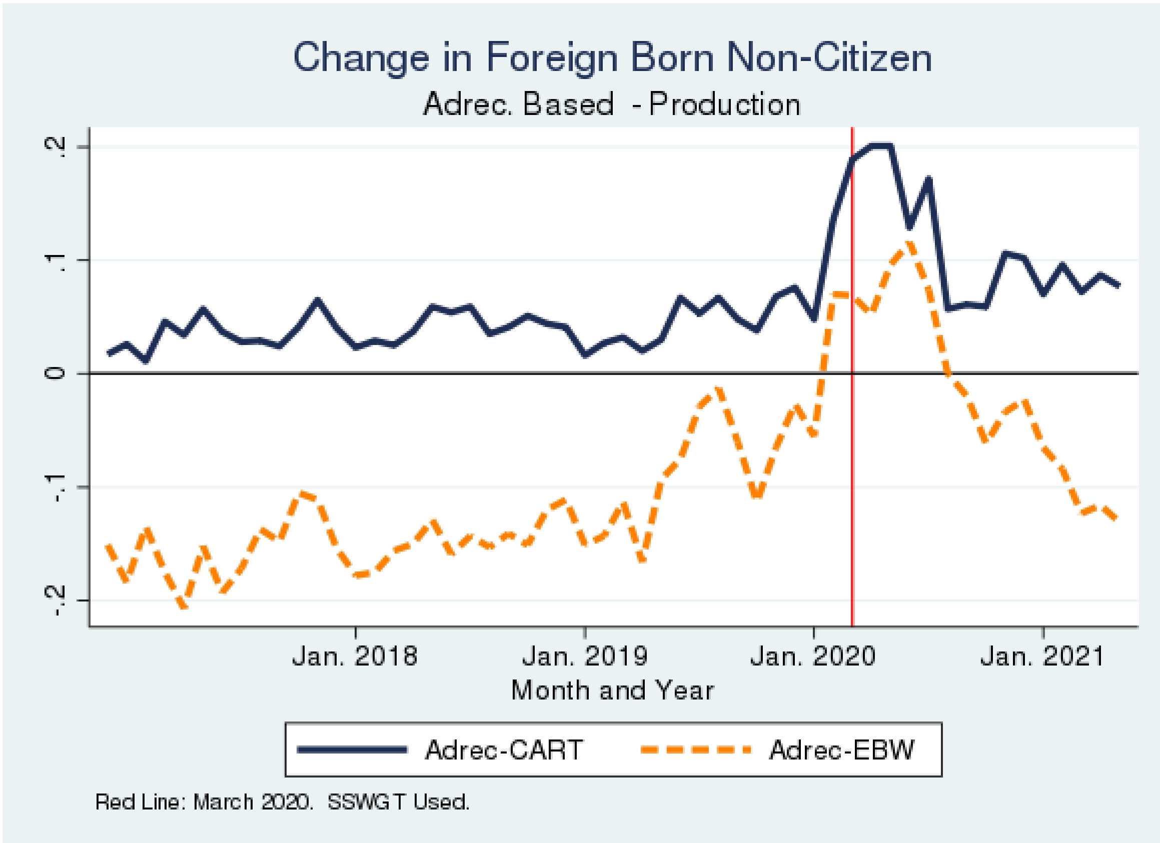


Figure present both levels (top panel) and the difference between the production estimate and the administrative data-based estimate (bottom panel). See Table 8 for more details.

Source: Basic Monthly CPS matched to administrative data